CropCode: Inverse Procedural Modeling of Occluded Row Crop Geometry from Sparse Multimodal Field Observations

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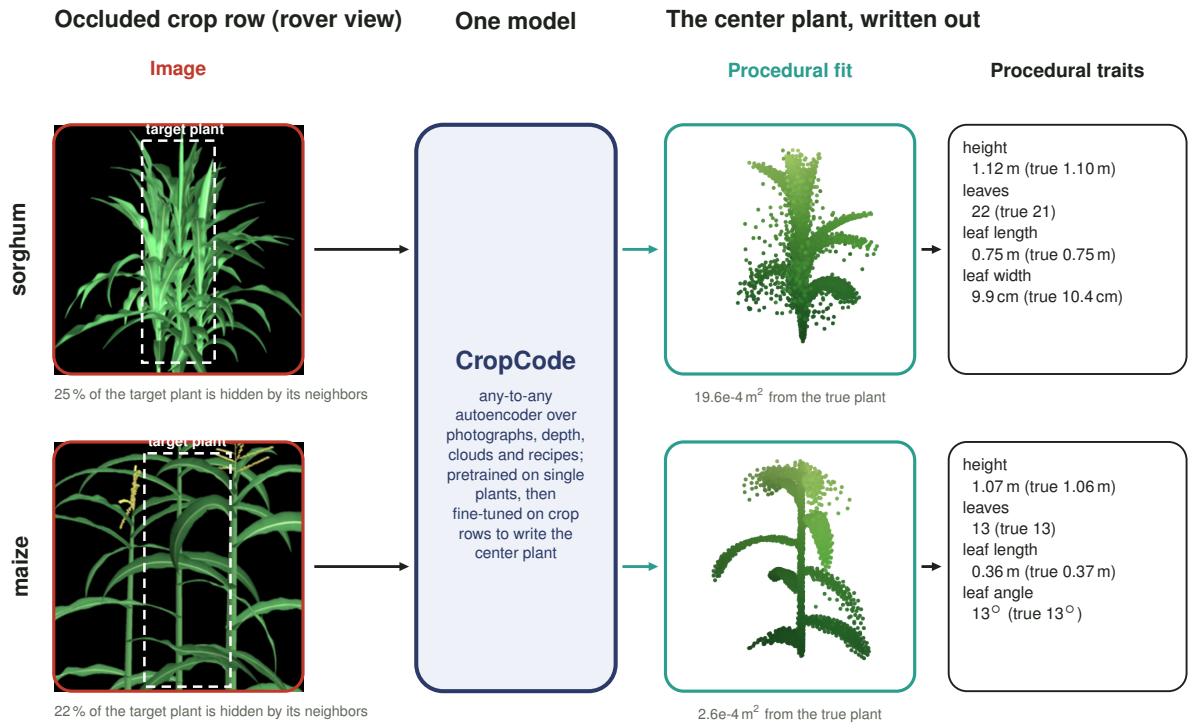


Figure 1. **Any sensor in, procedural plant out.** From any subset of field-practical modalities (a single photograph, a depth frame, or a point cloud truncated by row neighbors), CROPCODE recovers the full procedural representation of the plant: a kilobyte-scale, physiologically interpretable parameter set that can be re-rendered to complete geometry, used as simulation input, and used for trait analytics at field scale. Shown: one held-out simulated test row per species, seen broadside by the rover camera. The dashed box marks the target plant for the reader; the model receives no marking. The model fine-tuned on rows reads the row observation (here the photograph; a depth frame or point cloud serves equally) and writes the target plant's recipe. The *Procedural fit* panel shows the geometry the model writes for that plant (19.6 and 2.6×10^{-4} m^2 from the true plant), and the traits read from the recipe are shown beside the true values (Regime III, Sec. 4.4).

Abstract

Editable, biologically interpretable 3D models of field-grown crops support high-throughput phenotyping, in silico *breeding trials, and agronomic digital twins, but recovering such models from field sensor data remains difficult: plant organs are slender, neighboring plants occlude one another, and a field campaign rarely yields more than a single photograph, a noisy depth frame, or a partial point cloud per plant. Slender geometry under occlusion is a regime in which learned 3D completion and Chamfer-based evaluation both fail. In this work, we present* CROPCODE*, a multimodal masked-autoencoder framework that recovers the complete procedural representation of a plant, the generator parameters encoding leaf count, insertion heights, blade angles, and surface geometry, from any subset of RGB, depth, and point-*

cloud input in a single forward pass. The model is pretrained on 15,000 procedurally generated plants per species (maize and sorghum), fine-tuned on rows in which simulated neighbors occlude the target, and distilled from a full-modality teacher into single-sensor students. Distillation lowers the cross-modal generation loss by 30% in sorghum and 44% in maize, and halves the sorghum parameter error from a point cloud alone. Training on rows removes 92% (maize) and 85% (sorghum) of the error that neighbors add for a model trained on isolated plants. Controlled ablations in both species show that exposing most of the procedural parameters to the encoder degrades phenotype recovery, while using them only as a reconstruction target degrades the sorghum representation and improves the maize one. We further show that Chamfer distance and linear probes on the frozen representation rank input modalities differently on slender geometry, and we adopt the probe as the deciding metric. On crop rows built from real plants of both species, accuracy degrades zero-shot on plants taller than any training plant, and a brief fine-tune on 75% of the real plants brings the held-out rest to a 1.6% median height error and a leaf count within one leaf on 70% of maize stops, at milliseconds per plant against hours of manual specification or minutes of staged-pipeline computation. We will release FIELDSCAN*, the 30,000-plant, two-species benchmark behind these experiments, which pairs multimodal field captures under simulated row occlusion with exact procedural ground truth.*

1. Introduction

Accurate three-dimensional models of individual crop plants enable high-throughput phenotyping of morphological traits such as leaf angle, internode length, and canopy architecture, and they are the basis of emerging agricultural digital twins in which breeding trials run *in silico* [19, 25]. The representation best suited to these uses is procedural: a compact set of biologically interpretable parameters that a generator expands into complete, editable, simulation-ready geometry [20, 35]. A procedural plant occupies a few kilobytes where a laser scan occupies megabytes, every parameter corresponds to a trait used by agronomists, and the model can be edited, re-rendered, and perturbed in ways that unstructured geometry does not support. Field crops also instantiate a challenge of general interest in 3D computer vision: recovering a parametric model of a thin, heavily occluded object from partial multimodal observations.

Yet every existing route to procedural structure presupposes a rich, clean observation: manual specification takes hours per plant [20]; a staged pipeline annotates laser scans with a vision-language model and fits a descriptor to 5.4 mm median accuracy, but needs a complete scan of an isolated plant and about ten minutes per plant [18]; per-scene optimization searches generator parameters against multi-view imagery [52]; and learned models infer parametric plants only from a clean, complete point cloud [8, 14]. A phenotyping campaign produces the opposite: thousands of plants per day, each observed by whichever sensor was practical. Crop plants are also slender (a maize blade is centimeters wide, millimeters thick, up to a meter long) and occluded by their row neighbors, whose leaves interleave with the target: in our field simulations a third of the target's pixels and two thirds of the scene's point cloud belong to neighbors. This regime breaks the assumptions of learned 3D completion, whose benchmarks hold compact, largely convex objects [49, 51], and of Chamfer distance, which is structurally insensitive to thin sheets [42, 47]; occlusion-aware completion of plant organs outputs unstructured geometry rather than structure [27]. To our knowledge, no prior method recovers a procedural, re-renderable representation of slender crop organs from a single field-practical observation under row-neighbor occlusion.

CROPCODE makes this inverse problem tractable by using the procedural generator itself as the prior. We pretrain a four-stream masked autoencoder [3, 10, 21, 31] over RGB, depth, point clouds and tokenized procedural parameters of 15,000 generated plants per species (maize and sorghum), fine-tune it on simulated rows whose neighboring leaves bend on contact, and distill a full-modality teacher [22] into single-sensor students; random modality subsets are masked in pretraining, so one model completes any subset at inference. Distillation lowers the cross-modal generation loss by 30% in sorghum and 44% in maize and halves the sorghum parameter error from a point cloud alone; row fine-tuning removes 92% (maize) and 85% (sorghum) of the error the neighbors add. We also answer a design question that arises for any model with access to generative structure: a parameter stream largely visible to the encoder harms the representation in both species, while budget-matched controls show its value as a pure reconstruction target depends on the generator (it degrades sorghum and improves maize). Supervising parameters as a generation target still lets the distilled generalist beat a parameter-only specialist at parameter regression (recipe MAE from the point cloud 0.019 against 0.036), consistent with modality competition [34] and negative transfer [48] from a stream that already contains the answer. Finally, depth lowers Chamfer by 25–28% while a frozen-representation probe finds little gain from it and identifies RGB as the main source of trait information: on slender geometry the two metrics measure different quantities, and we adopt the probe as the deciding metric [6, 47].

We claim that fragmentary field sensing suffices to recover a complete, interpretable procedural plant, provided the generator is used for both the training data and the output representation. We test this claim on rows built from real maize and sorghum plants with independently reconstructed

ground truth [18, 25], in which neighbors occlude the target: zero-shot, accuracy degrades on plants outside the generator's size range, and fine-tuning on 75% of the real plants recovers the held-out plants' height to a 1.6% median error and their leaf count to within one leaf on 70% of maize stops, in milliseconds per plant. The key contributions of this work are:

1. **Task and framework.** The first formulation and system, to our knowledge, for recovering procedural plant representations of slender, row-occluded field crops from *any subset* of RGB, depth, and point-cloud input in one amortized forward pass, with full-modal to single-sensor distillation (generation loss -30% in sorghum and -44% in maize; sorghum parameter error from a point cloud alone halved), and row fine-tuning that removes most of the error added by neighboring plants (92% in maize, 85% in sorghum).
2. **A design rule for structured modalities.** Controlled, budget-matched, two-species evidence that a parameter stream largely visible to the encoder degrades the sensor representation in both species, while the value of the parameters as a reconstruction target depends on the generator (it degrades the sorghum representation and improves the maize one); we expect this guidance to transfer to any multimodal model with access to privileged generative structure.
3. **An evaluation protocol for slender geometry.** A demonstration that Chamfer distance and downstream phenotype probes disagree systematically on slender organs, and a frozen-latent probe protocol that resolves modality decisions where Chamfer is insensitive.
4. **A benchmark.** FIELDSCAN, a two-species procedural field benchmark of 30,000 plants spanning diverse phenotypes, generator configurations, and row occlusion severities, with paired multimodal captures, physically simulated row occlusion, and exact parameter ground truth, together with a real-field evaluation split carrying reconstructed procedural ground truth across the genotypes of a diversity panel [25]. No existing dataset combines these elements (Tab. 2) [41, 51]. We will release the data, generation code, and evaluation scripts.

2. Related Work

Appendix E reviews each of these literatures in detail; here we state what distinguishes our setting.

Recovering procedural plants. Tree-generator parameters have been recovered per instance by stochastic search [43], MCMC inference [44] and L-system induction [17, 28]; recent crop systems learn parametric morphology from clean point clouds [8, 14], fit procedural models against multi-view imagery [52], or reconstruct procedural maize from terrestrial laser scans through a staged vision-language pipeline [9, 18]. Each presupposes a single rich observation of an isolated plant; none addresses missing modalities or inter-plant occlusion.

Programs from observations. Shape programs [45], diffusion-based inverse procedural content generation [53] and CAD generation [24, 26] decode parametric programs from observations, but for rigid objects observed cleanly and in isolation, and none asks whether the program parameters should be visible to the encoder at all.

Completion and slender geometry. Learned point cloud completion [5, 49–51] trains on compact, largely convex benchmarks; completion of occluded plant organs [27] and thin-structure reconstruction [13, 38] output unstructured geometry rather than an editable model. Chamfer distance is structurally insensitive to thin sheets [30, 42, 47], so we evaluate with trait probes alongside geometric metrics (Sec. 4.7).

Multimodal masked pretraining and synthetic data. Masked autoencoding [21] extends to multiple modalities [3], any-to-any generation [4, 31], point clouds [33] and combined RGB, depth and point cloud inputs in EmbodiedMAE, on which our architecture is based [10]. We add a structured generative modality encoded with continuous number tokens [15] and distill a generative full-modality teacher [22] into single-sensor students, with missing-modality robustness a recognized concern [40]. Pretraining on purely synthetic data is viable from fractals to procedural scenes [16, 23, 39], with domain randomization reducing the gap [46] and task alignment setting its value [11]; our setting is the favorable case, since the generator defines the output representation and is paired with real field evaluation data [25], and our caveat is that exposing most of its parameters to the encoder reduces representation quality in both species (Sec. 4.6).

3. Method

We formulate the inverse problem (Sec. 3.1), build the FIELDSCAN data engine (Sec. 3.2), pretrain a four-stream masked autoencoder (Sec. 3.3), distill it into single-sensor students (Sec. 3.4) and define the evaluation protocols (Sec. 3.5); Fig. 2 gives the overview.

3.1. Problem formulation: inverse procedural reconstruction

A procedural generator G maps a parameter vector $\theta \in \mathbb{R}^{d_\theta}$ to an editable 3D mesh $G(\theta)$; θ describes the plant completely, from stem height and internode spacing to each leaf's insertion height, blade length and width, and insertion, roll and azimuth angles. Field sensors yield partial, noisy,

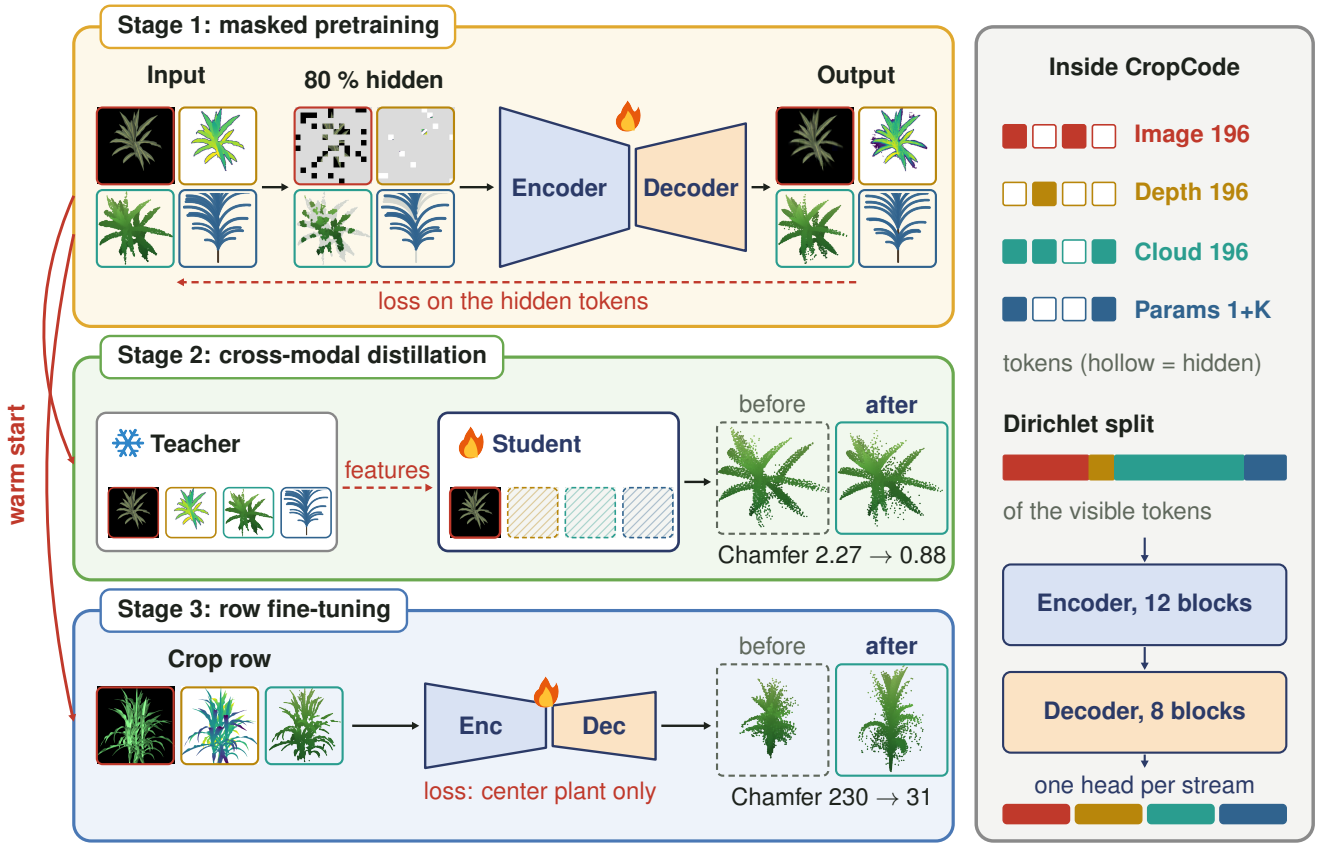


Figure 2. **CROPCODE in three training stages.** Stage 1: the four streams of a plant are tokenized, 80% of the tokens are hidden under a Dirichlet split across streams, and a shared encoder and decoder reconstruct every stream (image, depth and parameter losses on the hidden tokens; the point-cloud loss, QAL plus Sinkhorn, on the whole cloud). The procedural parameters are drawn as a stem-and-leaves schematic. Stage 2: a frozen teacher that reads all four streams passes its decoder features and latent to a student that reads one stream; shown is the point cloud generated from the image alone before and after distillation. Stage 3: fine-tuning on crop rows with the loss taken against the center plant alone; shown is the center plant written from the row before (no row fine-tuning) and after. Chamfer: sorghum test means from Table 8 ($\times 10^{-3}$) and Table 7 (all streams, $\times 10^{-4}$ m^2), both in Appendix A. Stages 2 and 3 start from the Stage 1 weights. Examples: one held-out sorghum test plant and row.

occluded observations: from the modality universe $\mathcal{M} = \{\mathrm{rgb}, \mathrm{d}, \mathrm{pc}\}$ a campaign acquires some subset $S \subseteq \mathcal{M}$, and row neighbors occlude the target. We learn an amortized feed-forward inverse

$$f_\phi(\{x^m : m \in S\}) = \left(\hat{\theta}, \{\hat{x}^m : m \in \mathcal{M}\}\right), \qquad (1)$$

returning the predicted recipe $\hat{\theta}$ and the completed streams. Where raw geometric completion can only guess hidden points, predicting $\hat{\theta}$ uses G as the prior: $G(\hat{\theta})$ is structurally complete, editable and interpretable in agronomic traits, and the inversion runs in one forward pass of milliseconds instead of minutes to hours of per-plant optimization.

3.2. A procedural field-data engine: FIELDSCAN

To train and evaluate CROPCODE, we construct FIELDSCAN, a synthetic benchmark pairing exact procedural ground-truth parameters with multimodal field observations under realistic row neighbor occlusion (Tab. 1). Figure 3 shows, for each species, the four streams of one view, the ten cameras of one plant and a row scene.

Generation in brief. Each species' generator draws 15,000 plants: maize 0.62–1.44 m tall with 3–22 leaves, its traits coupled through four latent factors so that height and leaf count stay weakly correlated ($r = 0.20$); sorghum (Evo-Engine) with stems of 0.19–1.74 m and 4–35 leaves, stem length tied to leaf count ($R^2 = 0.988$), so the two species cover a weakly and a strongly coupled morphological regime. Ten views per plant are rendered at 1024^2 (40° field of view) with photograph, losslessly packed metric depth and camera calibration: sorghum from a fixed Fibonacci rig at 2.5 m with the complete plant cloud per view, maize from random directions at a box-fitted distance with partial 8,192-point

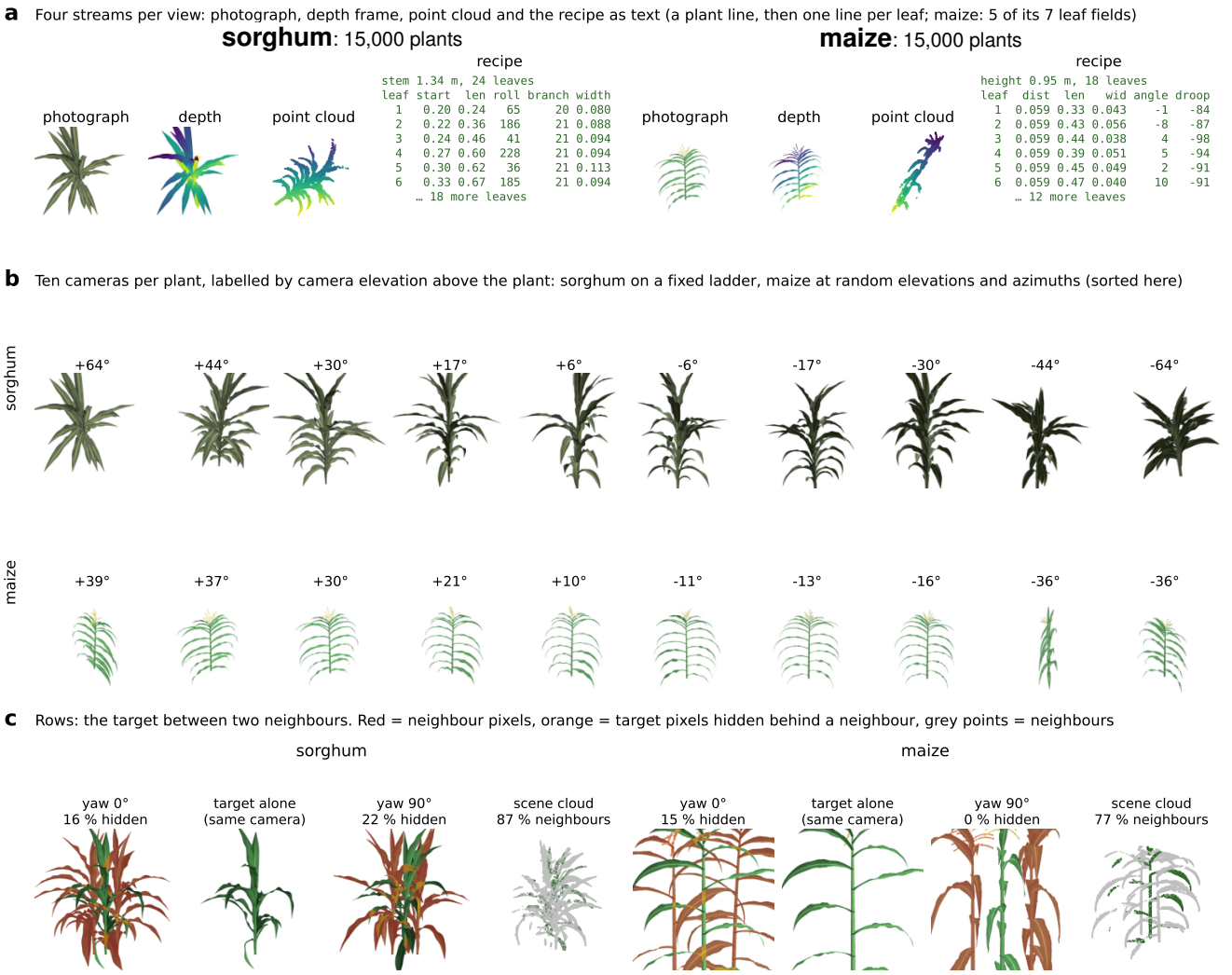


Figure 3. **FieldScan at a glance.** (a) The four streams of one view per species: photograph, depth frame, camera-frame point cloud and the procedural recipe as text (a plant line, then one line per leaf; maize shows five of its seven leaf fields). (b) The ten cameras of one plant, labelled by elevation above the plant: sorghum on a fixed ladder from $+64°$ to $-64°$, maize at random elevations and azimuths (sorted here). (c) Row scenes at yaw $0°$ and $90°$: neighbor pixels in red, target pixels hidden behind a neighbor in orange, the target plant alone from the same camera, and the scene cloud with neighbor points in grey.

clouds back-projected from depth. Row scenes place each target between two neighbors at 0.15 m spacing across ten yaws ($0°$–$90°$, $\pm 15°$ neighbor offsets); an articulated contact solver bends leaf blades (8 cm jointed segments, exact triangle–triangle tests) so that neighbors yield first and the target keeps its grown shape wherever a neighbor can give way, and a rover camera captures the scene (RGB, depth, 24,576-point cloud) with the bent target alone (8,192 points) as ground truth. Neighbors account for 59–68% of plant pixels in maize rows and hide 23% of the target's pixels in sorghum. Plants split 70/15/15 by plant, with validation and test enriched toward extreme morphologies by Mahalanobis ranking. Appendix I gives the generators, the renderer, the contact solver and the partitioning in full.

3.3. Four-stream masked pretraining

Multimodal tokenization. Building on Embodied-MAE [10], RGB and depth are split into 196 patches on a 14×14 grid and linearly projected to 768-d tokens with 2D sine–cosine positions; point clouds become 196 farthest-point-sampled groups of 32 neighbors with PointNet features [33, 36]; and the parameter vector θ becomes $1 + K$ continuous numerical tokens (one plant token plus K leaf slots, $K = 28$ for maize and 24 for sorghum), affinely

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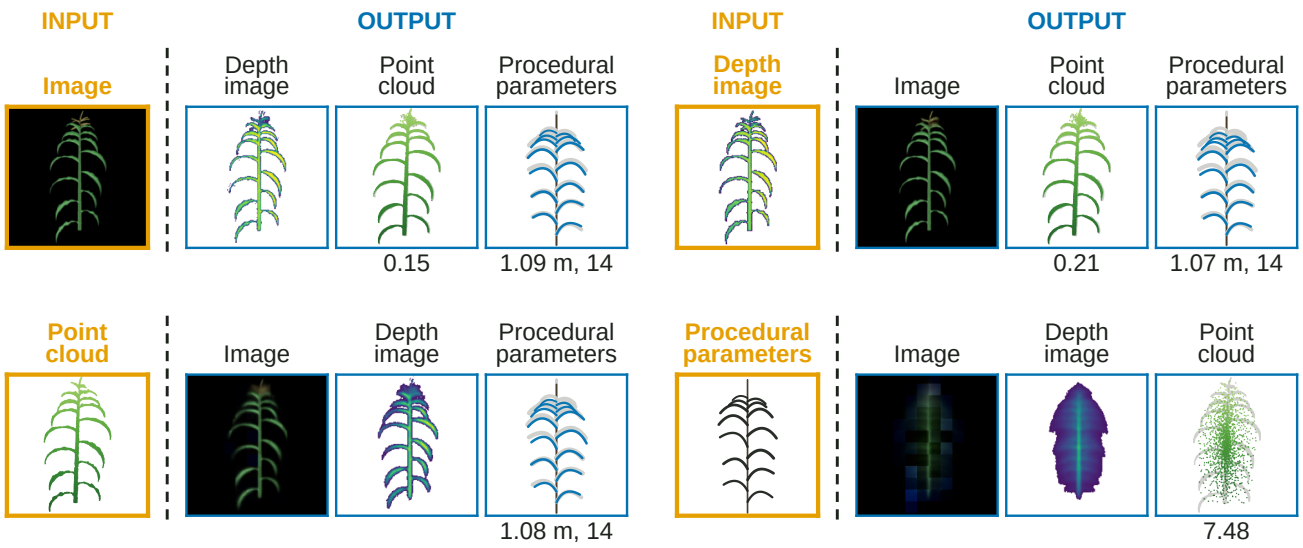


Figure 4. **Any-to-all, maize: each stream alone in, the other three out.** The distilled generalist on one held-out test plant. Each block gives it one stream (orange, left of the dashed line) and shows the three streams it generates (blue). Generated point clouds and procedural parameters are drawn over the ground truth in grey, the parameters as a stem-and-leaves schematic. Under each generated point cloud: its Chamfer distance (10^{-3}); under each generated parameter set: height and leaf count (truth 1.12 m, 14 leaves). The plant is selected for clarity among the test plants with the lowest image-to-cloud Chamfer, seen within 25° of the horizon. The parameters alone give only a coarse plant (Chamfer 7.48 here, 12.8 on average in Table 8).

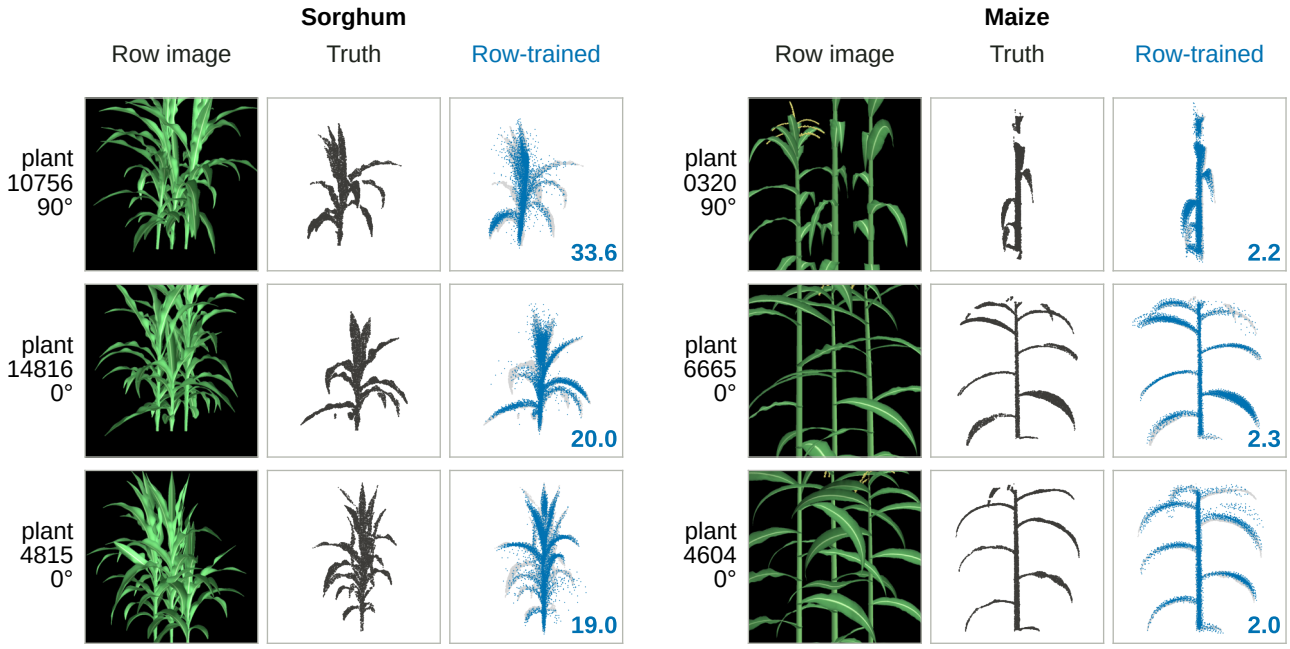


Figure 5. **Row occlusion reconstruction on held-out test plants.** Sorghum (left) and maize (right); 0° is broadside, 90° end-on. Columns: the row image the model reads (neighbors included), the center plant's true point cloud, and the point cloud written by the row-trained model (blue) over the truth in grey, with its Chamfer distance to the truth (10^{-4} m^2). The plants were selected for clarity among those where the row-trained model is in its best 35% and the isolated-trained control in its worst 40% at both yaws; Figs. 28 and 29 show ten more plants per species, spread over the whole error range.

normalized to $[0, 1]$ and embedded xVal-style [15]. Learned modality embeddings and a `[CLS]` token complete the

Parameter / Configuration	Maize	Sorghum
Total generated plants	15,000	15,000
Split (train / val / test)	10,500 / 2,250 / 2,250	
Views per plant	10 (random sphere)	10 (fixed rig)
Camera distance	box-fitted	2.5 m
Per-view point cloud	8,192 pts (visible)	complete plant cloud
Parameter dim. d_θ	$14{+}14{\times}28{=}406$	$9{+}5{\times}24{=}129$
Leaf count range	3–22	4–35
Plant height range	0.62–1.44 m	0.19–1.74 m (stem)
Row scenes × orientations	15,000 × 10	15,000 × 10
Row camera	level rover, 1.0 m	level rover, 2.5 m
Row cloud (input / target)	24,576 / 8,192 pts	
Real-field split	520 real plants, 2 layouts	30 scanned plants, 2 layouts

Table 1. **FieldScan dataset configuration.** Parameter dimensions include macro plant-level traits plus K structured leaf parameter slots. Every row scene pairs a target main plant with two adjacent row neighbors spaced at 0.15 m stem intervals.

Dataset	Real scans	Slender	Row occl.	RGB+D+PC	Exact proc. GT	Scale
PCN / ShapeNet-55 [49, 51]	×	×	×	×	×	55,000
Pheno4D [41]	✓	✓	×	×	×	84
Crops3D [54]	✓	✓	×	×	×	2,500
MaizeField3D [25]	✓	✓	×	×	(✓)	1,000
FieldScan (ours)	✓†	✓	✓	✓	✓	**30,000**

Table 2. **Dataset comparison.** Exact Procedural GT indicates exact ground-truth generative parameter vectors per plant; (✓) denotes parameters fitted post hoc via optimization. †FieldScan includes real-field evaluation scans; training splits consist of procedural synthetic data.

sequence for a 12-block ViT encoder (width 768) and an 8-block decoder (width 512), with linear RGB and depth heads, a FoldingNet cloud head (41 points per token) and an MLP for the parameters. Scored against ground truth, real leaf slots are selected by the validity mask; used without ground truth (Tab. 5, Appendix A), the leaf count is read from the recipe itself, which maize stores in the plant token and sorghum implies through the generator's stem-length rule.

Dynamic Dirichlet masking. A total masking budget of $\rho = 0.80$ leaves $V = \lfloor (1-\rho)L \rfloor$ visible tokens (122 of $L = 613$ for sorghum, 123 of 617 for maize), allocated across streams by a symmetric Dirichlet draw at every step, each stream keeping at least one visible and 25% masked tokens, so one network learns to complete any modality subset. The parameter stream sits inside this shared budget in the headline setup (∼58% visible on average); the control arms of Sec. 4.6 instead gate it at a fixed visibility ρ_θ, with $\rho_\theta = 1$ making the parameters a decoder-only reconstruction target.

Multi-term reconstruction objective. The objective sums modality losses,

$$\mathcal{L}_{\text{rec}} = \ell_{\text{rgb}} + \ell_{\text{d}} + \ell_{\text{pc}} + 5\ell_\theta. \tag{2}$$

with ℓ_{rgb} MSE on patch-normalized pixels [21], ℓ_{d} MSE on min–max normalized depth, and ℓ_θ Smooth-L_1 on masked parameter tokens. Standard Chamfer lets many predicted points match one target point, collapsing decoders onto stems [42, 47], so the cloud loss is sigmoid-weighted,

$$\ell_{\text{pc}} = \frac{1}{|\hat{P}|}\sum_{p\in\hat{P}} w(\delta(p,P))\,\delta(p,P) + \frac{1}{|P|}\sum_{q\in P} w(\delta(q,\hat{P}))\,\delta(q,\hat{P}), \tag{3}$$

with $\delta(p,P) = \min_{q\in P}\|p-q\|_2$, $w(r) = \sigma(a(r-\tau))$, $a = 100$ and $\tau = 0.01$ of the unit sphere, and the headline setup adds a debiased Sinkhorn divergence [12] that matches the clouds as distributions and so penalizes missing blades,

$$\ell^\star_{\text{pc}} = \ell_{\text{pc}} + 0.5\, S_\varepsilon\left(\hat{P}_{1024}, P_{1024}\right), \tag{4}$$

on random 1,024-point subsets ($\varepsilon = 0.02$). Models train with AdamW (cosine decay): the headline schedule peaks at 6×10^{-4} over 1,000 epochs of all ten views at global batch 256; ablation arms share a matched budget of 197,400 steps at global batch 32, one view per plant per epoch, peaking at 1.5×10^{-4}, with Eq. (3) alone as the cloud loss.

3.4. Full-modality teacher to single-sensor students

Single-sensor deployment shows the encoder a configuration pretraining never does: every token of the unobserved streams absent. A frozen four-stream teacher closes the gap, with the parameters as privileged input (every token visible for sorghum; half of each stream for maize, whose cloud reconstruction degrades when fully visible). It emits a global latent $c^T \in \mathbb{R}^{768}$ and normalized decoder features $F^T \in \mathbb{R}^{L\times512}$; students start from the teacher's weights, train 100 epochs (AdamW, 10^{-4}, global batch 128), receive a single source $s \in \{\text{rgb}, \text{d}, \text{pc}, \theta\}$ on 80% of steps and standard four-stream pretraining on the rest, and minimize

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{rec}} + \frac{\lambda_F}{|\mathcal{J}_{\neg s}|}\sum_{j\in\mathcal{J}_{\neg s}} \|F^S_j - F^T_j\|_2^2 + \lambda_c\|c^S - c^T\|_2^2, \tag{5}$$

where $\mathcal{J}_{\neg s}$ are the decoder positions of the unobserved non-source streams ($\lambda_F = 1.0$, $\lambda_c = 0.5$). At deployment each sensor is evaluated alone.

3.5. Field evaluation and transfer protocols

We evaluate in three regimes. **(1) Trait probing**: ridge probes on frozen `[CLS]` tokens, fit on the training plants and scored on the held-out splits with one deterministic view per plant and the parameter inputs zeroed against label leakage; a change counts as a win only when its 95% paired-bootstrap interval (2,000 resamples over plants) excludes zero on validation and test. Under partial clouds the probe is refit per visibility condition (*matched*) or applied unchanged from the complete-cloud fit (*transferred*). **(2)**

Rows: zero-shot transfer of isolated-trained models, and row fine-tuning that reconstructs the target plant behind its neighbors, with an isolated-control arm on the same schedule and an echo test that scores generated clouds against both the target and the input scene. **(3) Real-field validation** on MaizeField3D captures [25], against the procedural fits of a staged vision-language pipeline [18] as independently reconstructed ground truth.

4. Experiments

Unless a figure number five or lower is named, the figures referenced in this and the following sections are in Appendix B.

In this section, we present an extensive experimental evaluation of CROPCODE. We describe our experimental setup (Sec. 4.1), followed by evaluations across three deployment regimes: isolated single-view reconstruction (Sec. 4.2), difficulty progression across viewpoint and self-occlusion (Sec. 4.3), and neighbor occlusion in realistic field rows (Sec. 4.4). We then analyze cross-modal distillation (Sec. 4.5), investigate the placement of the procedural parameter stream (Sec. 4.6), uncover systematic divergences between geometric metrics and downstream trait probes (Sec. 4.7), describe the real-field validation protocol (Sec. 4.8), and analyze scaling, cross-species transfer, and latent space structure (Sec. 4.9).

4.1. Experimental setup

Models and compute budgets. All ablation arms share 197,400 optimizer steps at global batch 32 (600 epochs of one random view per plant), identical view-sampling seeds and a ViT-Base encoder (114 M parameters) unless capacity is varied; data-scale arms adjust their epoch counts to keep the step budget. The headline model trains for 1,000 epochs on all ten views at global batch 256 with the Sinkhorn term and initializes every teacher and student of Sec. 4.5. Results are on the validation split unless stated otherwise.

Evaluation metrics. Geometry is symmetric squared Chamfer in the unit-sphere frame ($\times 10^{-3}$; 1.0 corresponds to $\approx$ 33 mm RMS on the median sorghum plant and $\approx$ 19 mm on the median maize plant, with metric scales measured per plant), evaluated in the masked regime of pretraining for pretrained models and with unmasked inputs for distilled students. Parameter accuracy is MAE over normalized $[0, 1]$ fields of hidden, non-padding leaf tokens, reported as trait *skill* where informative,

$$\text{Skill} = 1 - \frac{\text{MAE}}{\text{MAE}_{\text{mean}}}, \tag{6}$$

so that 1 is perfect recovery and 0 matches predicting the dataset mean. Representation quality is the probe of Sec. 3.5, R^2 with 95% paired-bootstrap intervals on 2,250 validation and 2,250 test plants. Sorghum's four named developmental traits have effective rank two (height and leaf count couple at $r = 0.994$), so sorghum probing adds the generator's independent axes (leaf roll, blade length); maize's eleven traits are largely independent.

4.2. Regime I: single-view reconstruction of isolated plants

Table 5 (Appendix A) covers deployment on an isolated plant: one photograph, one depth frame or one point cloud in, the recipe and the other streams out in a single forward pass. Depth is the best single input for the cloud in both species (Chamfer 0.93 in sorghum and 0.64×10^{-3} in maize, a median of 30 and 12 mm RMS per plant, F_1 0.44 and 0.53), with the photograph within a factor of 1.5 of it; on the full test split, photograph-to-cloud Chamfer corresponds to a skill of 0.98–0.99 against the distance between two random plants (Sec. 4.5). The recipe is recovered from any single sensor: leaf count within one leaf for 91–100% of plants, height within 2–5%, and mean leaf-parameter skill 0.66–0.68 in sorghum and 0.57–0.59 in maize, high on the leaf's stem position and angle, moderate on length, and lowest on sorghum blade width, whose between-plant spread is small. The headline model given all three streams is no better than the generalist given one, because it was never trained to read every token of its input; the first release's three unrecoverable waviness fields are absent from the final generator. Averaging the recipes of three views improves every recipe column, while reading the recipe from a generated cloud loses almost all of it, so the recipe is read from the sensor directly (Tab. 5). Figure 6 (Appendix B) shows single-photograph reconstructions: the distilled generalist resolves individual leaf blades where the undistilled model generates diffuse clouds, and the procedural output can be edited and re-rendered.

4.3. Regime II: viewpoint elevation, self-occlusion, and complexity

Viewpoint elevation. Every view of a plant shares the same ground truth, so the elevation ladder isolates the sensor input. Photograph-to-cloud error follows a symmetric U over camera elevation, lowest side-on and highest looking steeply down or up; elevation explains $\eta^2 = 0.24$–0.31 of per-plant error variance while morphology correlates negligibly ($|\rho| \leq 0.06$), and distillation cuts the error by 28% across views without flattening the curve (Fig. 7A).

Training view distribution. Pretraining restricted to elevation bands (top-down $i \in \{0, 1\}$, horizontal $i \in \{4, 5\}$, or the single view 0) wins only on the trained elevations and loses elsewhere: the sorghum per-view probe drops by 0.066–0.144 on distant views while the all-view arm holds $R^2 = 0.734$–0.778 (Fig. 8), and in maize, probed at a random camera, every restricted arm loses 10–11 of 11 targets

	Sorghum					Maize			
Frozen Encoder	Height	Leaf Cnt	Branch	Roll	Leaf Len	Height	Leaf Angle	Curl	Stem Rad
DINOv2 ViT-B/14 (RGB) [32]	0.924	0.928	0.840	0.573	0.235	0.561	0.816	0.072	0.588
MultiMAE-B (RGB+D) [3]	0.922	0.925	0.833	0.679	0.232	0.570	0.783	0.069	0.442
Point-MAE (PC) [33]	0.958	0.963	0.863	0.716	0.188	0.509	0.863	0.199	0.426
EmbodiedMAE-B (RGB+D+PC) [10]	0.976	0.979	0.871	0.765	0.265	0.528	0.823	0.139	0.428
CROPCODE Headline (1,000 ep)	**0.983**	**0.988**	**0.912**	**0.798**	**0.371**	**0.747**	**0.967**	**0.739**	**0.841**

Table 3. **Comparison against published pre-trained encoders (E8).** Published self-supervised visual encoders are evaluated on identical held-out validation plants using mean-pooled features and the same ridge regression probe protocol. Every row is mean-pooled, including the headline model of each species at epoch 1,000. Values are validation R^2 (higher is better); bold marks the best value in each column.

(a) Modality value-add

	Chamfer	Frozen-Probe R^2 (Validation Set)					
Stream Set	$\times 10^{-3}$	Height	Leaf Cnt	Biomass	Branch	Roll	Leaf Len
Sorghum							
PC	5.00	0.895	0.896	0.892	0.792	0.393	0.088
PC + D	2.45	0.976	0.979	0.982	0.903	0.798	0.307
PC + RGB	2.57	**0.979**	**0.983**	**0.986**	**0.934**	0.808	0.267
PC + RGB + D	**1.85**	0.977	0.980	0.982	0.926	0.781	**0.309**
PC + θ	4.84	0.860	0.864	0.863	0.780	0.481	0.078
PC + D + θ	4.08	0.907	0.909	0.916	0.856	0.694	0.191
PC + RGB + θ	3.00	0.962	0.964	0.967	0.890	**0.810**	0.225
PC + RGB + D + θ	2.53	0.962	0.964	0.965	0.880	0.760	0.249
		Height	Leaf Cnt	Biomass	Branch	Curl	Stem Rad
Maize							
PC	2.63	0.171	0.670	0.610	0.756	0.114	0.289
PC + D	2.55	0.453	0.773	0.683	0.804	0.082	0.550
PC + RGB	1.61	0.620	**0.914**	**0.783**	**0.927**	0.334	0.753
PC + RGB + D	1.21	0.633	**0.914**	0.774	0.926	0.236	0.760
PC + θ	2.49	0.256	0.740	0.662	0.787	0.114	0.363
PC + D + θ	2.54	0.382	0.801	0.670	0.795	0.136	0.408
PC + RGB + θ	1.21	0.509	0.881	0.780	0.912	**0.371**	**0.763**
PC + RGB + D + θ	**0.99**	**0.635**	0.908	0.781	0.919	0.304	0.743

(b) Training dataset scale and encoder capacity

Dataset / Capacity Variant	Epochs	Chamfer	Height	Roll	Leaf Len
Sorghum: Training Dataset Scale (Base Model)					
1,000 Plants	6,169	3.13	0.952	0.716	0.249
3,000 Plants	2,100	2.86	0.932	0.679	0.213
10,000 Plants	631	2.48	0.962	0.749	0.245
10,500 Plants	600	2.53	0.962	0.760	0.249
Sorghum: Encoder Capacity (Full Dataset)					
ViT-Small (25 M)	600	2.86	0.964	**0.817**	**0.277**
ViT-Base (114 M)	600	**2.53**	0.962	0.760	0.249
ViT-Large (332 M)	600	2.68	0.948	0.711	0.246
		Chamfer	Height	Branch	Stem Rad
Maize: Training Dataset Scale (Base Model)					
1,000 Plants	6,169	3.54	0.393	0.851	0.593
3,000 Plants	2,100	1.46	0.524	0.867	0.624
10,000 Plants	631	0.87	0.427	0.853	0.717
10,500 Plants	600	0.99	0.635	0.919	0.743
Maize: Encoder Capacity (Full Dataset)					
ViT-Small (25 M)	600	1.22	0.563	0.886	0.682
ViT-Base (114 M)	600	**0.99**	**0.635**	**0.919**	**0.743**
ViT-Large (332 M)	600	1.06	0.580	0.887	0.614

Table 4. **Ablations at the shared step budget (197,400 optimizer steps). (a)** Modality value-add: models differing only in active input token streams; θ is the parameter stream as an encoder input under the shared Dirichlet budget. **(b)** Dataset scale and encoder capacity: at fixed step compute, larger encoders do not help and more data helps reconstruction more reliably than it helps the probes. Chamfer $\times 10^{-3}$ and linear-probe R^2 (validation).

under paired bootstraps, with the point-cloud F_1 agreeing. Covering the evaluation elevations matters more than the number of images; Appendix F.1 gives the per-arm numbers.

Self-occlusion under partial camera coverage. Clipping point clouds by mesh ray-casting to the surface visible from 1 or 3 cameras (49% and 78% coverage) leaves trait information intact when the probe is refit (height R^2 0.960 at one camera for the headline model) but breaks a probe transferred from complete clouds ($R^2 \rightarrow -0.26$; Fig. 9, Tab. 6, Appendix A). Partial visibility is a covariate shift in the latent space: field readouts must be refit on sensor-matched coverage.

Morphological complexity. After distillation, reconstruction error is flat across leaf-count deciles (Spearman $\rho = -0.07$ in sorghum, $+0.04$ in maize; Fig. 10).

4.4. Regime III: neighbor occlusion in realistic crop rows

Table 7 (Appendix A) compares identical plants in isolation and inside rows. Models trained on isolated plants fail under row occlusion, with Chamfer rising sixteenfold on maize ($1.8 \rightarrow 29.0 \times 10^{-4}$ m^2) and tenfold on sorghum ($13.1 \rightarrow 133.8$); the echo test shows they reconstruct neighboring foliage as the target. Row fine-tuning removes 92% of the neighbor-induced error in maize ($29.0 \rightarrow 4.0$) and 85% in sorghum ($133.8 \rightarrow 30.7$), isolating the target behind its neighbors (Fig. 5). Error peaks broadside and falls end-on in maize, whose leaf planes align with the row, and is yaw-independent in sorghum, whose headings are random (Fig. 11, Appendix B).

4.5. Any-subset modality reconstruction and distillation

Table 8 (Appendix A) reports cross-modal generation before and after distillation, and Figs. 4 and 12 show the generalist's outputs. Distillation cuts the mean generation loss by 30.2% in sorghum and 43.5% in maize: photograph-to-cloud Chamfer falls 61% ($2.27 \rightarrow 0.88 \times 10^{-3}$), photograph parameter MAE 76% ($0.058 \rightarrow 0.014$), and point-cloud parameter

MAE halves ($0.040 \rightarrow 0.020$). Sensor-plus-recipe pairs reach the lowest cloud errors (0.36–0.44×10^{-3}), and one student trained on all three input pairs matches the dedicated two-stream students within 0.03 in both species.

4.6. Influence of procedural parameter stream placement

Tables 4 and 9 (Appendix A) vary the active streams and the parameter stream's visibility. The image streams carry the trait information: adding RGB or depth to point clouds lifts sorghum height R^2 from 0.895 to 0.979 and 0.976, and RGB lifts maize from 0.171 to 0.620, while depth's 25–28% Chamfer gain leaves the probes nearly unchanged. Parameter tokens visible to the encoder state the targets outright, and the budget-matched controls separate the mechanisms: with the sensor-token budget restored, sorghum still loses on every target at any parameter visibility, including target-only supervision (0.966 against 0.977 on height), whereas maize improves on all eleven traits with the parameters gated to 20% or used only as a reconstruction target (0.776 against 0.633 on height). One interpretation is an informational shortcut that starves the sensor streams, with the decoder-target effect set by the generator's parameter structure (rank two in sorghum, eleven independent dimensions in maize); Figs. 13 and 14 (Appendix B) plot both ablations.

4.7. Divergence between Chamfer distance and probing metrics

Across twelve model variants the two metrics correlate overall (Spearman $\rho = -0.57$ to -0.88) but part exactly on design decisions (Fig. 15): adding depth cuts Chamfer by 25–28% yet leaves the probe unchanged, and parameter tokens lower maize Chamfer while degrading the sorghum probe. On slender organs, points clustered on the stalk minimize nearest-neighbor distance while whole blades go missing: precision exceeds recall for every variant (Fig. 16), and the loss ablation (Appendix F.13) confirms the training-side diagnosis, with QAL placing points precisely but covering little, the Sinkhorn divergence covering the plant but placing points poorly, and their combination, the headline loss, best at both thresholds. We therefore read the probe for modality decisions and train with the distribution-matching term (Eq. 4).

4.8. Real-world field validation

We test the row models on crop rows assembled from real plants: procedural models of 520 real maize plants and 30 scanned real sorghum plants (2.2–5.3 m tall), each repeated along a row at 15–25 cm spacing in two layouts (leaf planes parallel and perpendicular to the row), with a level camera stopping 18 times per row — 9,360 stops per layout for maize, 540 for sorghum. Inputs follow the Regime III conventions; the recipe targets are height and leaf count (the files carry no per-leaf recipe), and the renders keep the sensor synthetic, so real illumination and noise are not part of this test. Zero-shot, the row fine-tune is the best of the three Sec. 4.4 models on both maize layouts (Tab. 10, Appendix A; non-overlapping 95% intervals on Chamfer, best height and leaf count), so row training carries over to real geometry — but absolute errors sit an order of magnitude above the synthetic rows, mostly from plant size: 83% of the real maize plants are taller than any training plant, no model predicts above 1.24 m, and on the 87 in-range plants the median height error is 14% against 38% beyond. On sorghum, whose scans dwarf the training range, no model transfers zero-shot.

Fine-tuning the row model on 75% of the real plants of each species (split by plant; the real files supervise the center plant's cloud, depth and, for maize, height and leaf count) for 15 epochs on one GPU closes most of the gap on the held-out 25%: maize Chamfer falls from 59.5 to 19.7×10^{-4} m^2, median height error from 36% to 1.6%, leaf count within ± 1 rises from 45% to 70%, and sorghum Chamfer falls from 2,113 to 79 (eight test plants, indicative). The gain holds across inputs, layouts and occlusion quartiles (Tab. 11, Appendix A; Figs. 18, 19 and 17, Appendix B; Appendix G shows 20 more held-out plants). A few hundred real maize plants, or 19 real sorghum plants, suffice to close most of the generator-to-real gap while the representation and the procedural output stay unchanged; reading the center plant takes 10.6 ms per stop.

4.9. Representation quality, scaling, and transfer

Comparison against published encoders. Table 3 applies the trait probe of Sec. 3.5 to every encoder with mean-pooled features and only the streams each encoder was built for; Appendix C gives the full protocol, whose extreme-enriched splits make the R^2 values comparable within the table but not with other work. Three-stream CropCode (PC + RGB + D) outperforms DINOv2, MultiMAE, Point-MAE, and EmbodiedMAE across 8 of 9 sorghum traits and 10 of 11 maize traits. Under mean pooling, the headline model of each species is the strongest frozen encoder in the table: in sorghum it leads on size and leaf length (height 0.983, leaf length 0.371), and in maize it beats every published encoder on all eleven traits. Our architecture trained end to end on the trait labels, probed the same way, stays ahead of the maize headline model on height (0.785 against 0.747) and tassel droop and behind it on six of eleven traits (leaf angle 0.967 against 0.935, curl 0.739 against 0.618); full-pose Point-MAE looks strong on maize only through a renderer artifact (canonical leaf planes leak azimuth), and its height R^2 falls from 0.720 to 0.588 with realistic gravity-only poses while CropCode is unaffected (Appendix C). Appendix F.9 compares masked reconstruction with the encoders that have a decoder.

Scaling, transfer, and the latent space. Scaling dataset volume improves Chamfer distance monotonically (sorghum $3.13 \rightarrow 2.53 \times 10^{-3}$; maize $3.54 \rightarrow 0.99 \times 10^{-3}$), while probe scores are not monotone in maize (height $0.393 \rightarrow 0.635$ with dips at intermediate scales). Scaling model capacity from ViT-Base (114M) to ViT-Large (332M) degrades representation quality under fixed compute steps (Tab. 4, Fig. 20). Pretraining on one species and fine-tuning 150 epochs on the other beats training from scratch on every probed target (warm-started from the maize headline model, the sorghum three-stream arm wins all nine traits on both splits, validation Chamfer 1.83 against 3.38×10^{-3}), though decoders do not transfer zero-shot. Frozen `[CLS]` embeddings form continuous, smooth manifolds structured by global plant scale and organ posture rather than discrete clusters (Fig. 21), with a low effective participation ratio (4.4–6.3 of 768 dimensions in sorghum, 6.7–11.3 in maize).

5. Discussion and Limitations

Appendix D discusses each of the following points in full.

A design rule for privileged parameters. The parameter stream is the only modality that states the target traits verbatim, and it is absent in the field. Exposing it to the encoder degrades the sensor representation in both species, consistent with shortcut learning, modality competition [34] and negative gradient transfer [48]; the budget-matched controls (Tab. 9, Appendix A) show that even as a decoder-only target the stream helps maize (eleven largely independent dimensions) and hurts sorghum (effective rank two). Privileged generative parameters should never be largely visible to the encoder; whether they help as a pure reconstruction target depends on the generator.

Metric divergence on slender geometry. Chamfer distance rewards surface proximity and can be lowered by clustering points on stems while whole blades are missed; the frozen-latent probe measures recovered structure. The two metrics rank depth inclusion, data scaling and partial visibility differently (Sec. 4.7), so we report trait probes alongside geometric metrics and train with a distribution-matching Sinkhorn term (Eq. 4).

Partial sensing and rows. Camera elevation dominates single-image difficulty ($\eta^2 = 0.24$–0.31); phenotype information survives partial coverage but its readout shifts, so field probes must be refit on sensor-matched inputs; row fine-tuning removes most neighbor-induced error (92% in maize, 85% in sorghum); and distillation concentrates its gains on the weakest inputs without flattening the elevation difficulty curve.

Limitations. The ablations run one seed per arm at a shared step budget (run-to-run spread $\approx$ 2% in Chamfer, ≤ 0.011 in probe R^2); recovered representations are bounded by the generator's expressivity (disease, breakage and unvaried tassel droop lie outside it); and synthetic renderers can hand pose-aware baselines canonical cues, so the real-field evaluation (Sec. 4.8) remains the arbiter of transfer.

6. Conclusion

CROPCODE recovers editable, biologically interpretable procedural plant models from fragmentary, occluded field observations by using the procedural generator as both the training data engine and the output representation: any subset of photograph, depth and point cloud becomes a compact, simulation-ready parameter set in milliseconds. Distillation improves single-sensor parameter regression, row fine-tuning removes 85–92% of neighbor-induced error, and on rows built from real plants a brief fine-tune on a fraction of the plants recovers held-out height to 1.6% median error. Two design rules generalize beyond crops: privileged generative structure should not be largely visible to the encoder, and nearest-neighbor metrics are structurally insensitive to missing slender organs, so downstream probes should accompany them. We release FIELDSCAN, 30,000 plants across two species with multimodal captures, simulated row occlusion, exact procedural ground truth and a real-field evaluation split.

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A. Result Tables

The per-regime result tables referenced from the main text: single-view reconstruction (Tab. 5), partial camera coverage (Tab. 6), row occlusion (Tab. 7), any-subset generation and distillation (Tab. 8), the token-budget controls (Tab. 9) and the real-plant evaluation (Tabs. 10 and 11).

B. Result Figures

The result figures referenced from the main text, in the order of their sections.

C. Encoder-Comparison Probe Protocol

Probe protocol for the encoder comparison. Table 3 applies the trait probe of Sec. 3.5 to every encoder in the same way, with two changes that let the published encoders take part. First, each plant is represented by the mean of the encoder's output tokens, the one readout that every encoder supports; our models leave their `[CLS]` token out of this mean. Second, each encoder receives only the streams it was built for, all taken from view 00 of the plant: DINOv2 the photograph, MultiMAE the photograph and depth frame, Point-MAE the point cloud, EmbodiedMAE all three, and each CROPCODE encoder the sensor streams it was pretrained on. The procedural parameters are never an input. For each encoder and trait, the features of the 10,500 training plants are standardized with training statistics only. A ridge regression is then fitted, with its regularization strength chosen from $\{0.1, 1, 10, 10^2, 10^3, 10^4\}$ by five-fold cross-validation on the training plants. The table reports R^2 on the 2,250 validation plants. The network trained from scratch on the labels goes through the same probe on its frozen features; its own regression head is not used. The traits are the generator's ground truth for each plant. In sorghum they are stem length, leaf count, mean leaf branching angle, mean leaf roll and mean leaf length. In maize they are summed stem internode length, mean leaf angle, mean leaf curvature and stem radius. The validation and test splits are enriched with extreme plants, with about 1.8 times the training variance in sorghum and 1.5 to 2.6 times in maize. The R^2 values therefore rank the encoders within the table, but they are not comparable with values reported in other work. Test R^2 stays close to validation; for the maize headline model, height reaches 0.747 on validation and 0.759 on test.

Camera pose artifact analysis. Evaluating Point-MAE with full camera pose matrices yields strong maize performance. However, this is largely a synthetic renderer artifact: synthetic maize renders place leaf planes along a canonical world plane, so camera pose matrices leak plant azimuth orientation. Evaluating Point-MAE with gravity-only camera poses (realistic field calibration) reduces performance (R^2 drops from 0.720 to 0.588 on height), while CROPCODE performance remains unaffected.

D. Extended Discussion

The discussion of Sec. 5, in full.

We discuss the core insights, metric divergences, architectural principles, and limitations established by our experiments.

Privileged parameter visibility and modality competition. A central design question in multimodal architectures trained alongside generative models is where privileged structural information should enter the network. The procedural parameter stream θ is the only modality that states target phenotypic traits verbatim, yet it is never available at deployment in the field. Across our controlled ablations (Sec. 4.6), exposing parameter tokens to the encoder during pretraining degraded the quality of learned sensor representations in both sorghum and maize. In sorghum, every model arm that saw parameter tokens in the encoder learned inferior visual features, with the degradation on the size traits growing with the stream's visibility (Tab. 9).

This behavior can be understood through the lenses of shortcut learning, modality competition [34], and negative gradient transfer [48]. When ground-truth parameter tokens are visible to the transformer encoder, they directly encode target geometry. A plausible account is that the encoder learns a low-cost shortcut through these parameter tokens, starving the visual (RGB, depth) and point cloud streams of gradient updates; when parameter tokens are then absent at inference, the sensor representations prove impoverished.

The species-specific control arms (Tab. 9) reveal an additional nuance:

- **Sorghum**: Keeping parameter tokens strictly as a decoder target ($\rho_\theta = 1.0$) continues to lower sensor latent quality (R^2 drops from 0.977 to 0.966 on height). Sorghum morphological parameters are tightly coupled (effective rank two); a plausible mechanism is gradient competition during multi-task decoding that pulls sensor representations toward global plant size, the leading principal component of the sorghum latent.
- **Maize**: Keeping parameters strictly as a decoder target ($\rho_\theta = 1.0$) or restricting visibility to 20% improves sensor latent probing across all eleven traits ($R^2 = 0.776$ vs. 0.633 on height). Maize parameters contain 11 independent morphological dimensions; supervising them as a generation target provides structured multi-task decoder gradients, consistent with a regularizing effect on sensor latents without encoder shortcuts.

These findings yield an architectural design rule for multimodal masked autoencoders trained on synthetic or procedural data: *privileged generative parameters should never*

Input Modality	Leaf Count (Acc. ± 1) ↑	Height Error (Rel. Err.) ↓	Leaf Parameters (Mean Skill) ↑	F_1-score ($\tau = 0.01$) ↑	Chamfer $\times 10^{-3}$ ↓	RMS Error (mm) ↓
Sorghum						
RGB	0.909	0.036	0.656	0.386	1.36	37.1
Depth	0.956	0.026	0.679	**0.440**	**0.93**	**30.3**
Point Cloud	**0.998**	0.022	0.675	0.370	1.45	38.7
RGB + D + PC (headline)†	0.889	0.038	0.636	0.392	1.36	37.0
RGB, three views (recipes averaged)	0.995	**0.018**	**0.747**	–	–	–
Two-stage: RGB → cloud → recipe	0.301	0.221	−0.010	0.386	1.36	37.1
Maize						
RGB	0.973	0.045	0.567	0.482	0.83	13.6
Depth	0.972	0.046	0.571	**0.529**	**0.64**	**12.4**
Point Cloud	0.986	**0.043**	0.587	0.455	0.87	15.0
RGB + D + PC (headline)†	0.953	0.045	0.552	0.445	0.93	15.4
RGB, three views (recipes averaged)	**0.988**	0.044	**0.597**	–	–	–
Two-stage: RGB → cloud → recipe	0.066	0.131	0.015	0.482	0.83	13.6

Table 5. **Regime I: single-view reconstruction on isolated plants.** Each row is the distilled generalist given that input alone, on view 00 of every validation plant (2,250 per species). †The headline model given all three sensor streams, scored in its masked pretraining regime (80% of tokens hidden); with every token visible its cloud collapses. Leaf count and height are read from the predicted recipe: maize stores the leaf count in the plant token, sorghum does not, so its count is derived from the predicted stem length through the generator's rule (stem length $= 0.05n - 0.001$). Leaf-parameter skill is Eq. (6) averaged over leaf fields on real leaf slots. Chamfer and F_1 against the complete plant cloud, unit-sphere frame; the distillation history of Tab. 8 averages over random views, so its Chamfer differs. The three-view row averages the recipes predicted from the photographs of views 00, 01 and 02 (recipe columns only: each view's cloud is in its own camera frame). The two-stage row generates the cloud from the view-00 photograph and reads the recipe from that generated cloud; its cloud columns are the RGB row's. Reading the recipe from a generated cloud loses almost all of it (sorghum leaf skill 0.66 to −0.01), so the recipe is read from the sensor directly. Leaf-count accuracy is the fraction of the 2,250 plants whose predicted count is within one leaf of the true count; height error is the median relative error $|\hat{h} - h|/h$ over the same plants. Chamfer is the symmetric squared distance in the unit-sphere frame ($\times 10^{-3}$). The RMS column converts it to millimetres per plant (its square root times the plant's metric scale, the metres per unit-sphere unit measured on the raw camera-frame cloud; median 1.03 m in sorghum, 0.61 m in maize) and reports the median over plants. The rule above the two-stage row separates the end-to-end model, which reads the recipe from the sensor, from this modular alternative.

	Matched Probe (R^2, Height)			Transferred Probe (R^2, Height)			Chamfer ($\times 10^{-3}$)		
Input Modality Cloud Coverage	Full 100%	3-Cam 78%	1-Cam 49%	Full 100%	3-Cam 78%	1-Cam 49%	Full 100%	3-Cam 78%	1-Cam 49%
Point Cloud (PC)	0.888	0.839	0.765	0.888	0.509	0.575	4.99	6.18	12.8
PC + RGB	0.977	0.967	0.948	0.977	0.624	0.001	2.49	3.64	11.5
PC + RGB + D	0.975	0.968	0.956	0.975	0.721	0.576	1.76	3.06	11.4
Four-Stream Model	0.961	0.953	0.946	0.961	0.748	0.818	2.52	3.62	11.4
Headline Teacher	**0.982**	**0.975**	**0.960**	**0.982**	0.623	−0.26	**1.36**	3.14	11.6
Maize: Full = 10 cameras fused; coverage 100 / 96 / 72% of the 10-camera cloud									
Headline Teacher	0.719	0.698	0.754	0.108	0.251	0.754	4.98	4.54	1.15

Table 6. **Self-occlusion under partial camera coverage (sorghum test).** Evaluated on point clouds restricted to surface points visible from 1 or 3 camera perspectives via mesh ray-casting. Matched: linear probe refit per visibility condition. Transferred: linear probe trained on complete clouds applied directly without fine-tuning. Chamfer distance is evaluated against the complete ground-truth plant point cloud. Maize clouds are single-camera captures, so the maize model trained on the 1-camera condition; its transferred probe is fitted there and applied to the fused clouds, and its Chamfer is against the 10-camera cloud (val).

be largely visible to the encoder during pretraining. Setting parameter tokens strictly as a decoder reconstruction target ($\rho_\theta = 1.0$) or applying heavy gating ($\rho_\theta \leq 0.20$) prevents encoder shortcut learning while preserving generative parameter output capabilities; whether the parameters help even as a pure reconstruction target depends on the generator.

Metric divergence on slender geometry. A critical empirical finding of this work is the structural divergence between nearest-neighbor point cloud metrics (such as Chamfer distance) and downstream phenotypic trait probing on slender geometry (Sec. 4.7, Fig. 15). Standard Chamfer distance evaluates surface proximity by computing average nearest-neighbor point displacements:

$$\mathrm{CD}(P, \hat{P}) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in \hat{P}} \|p - q\|_2^2 + \frac{1}{|\hat{P}|} \sum_{q \in \hat{P}} \min_{p \in P} \|p - q\|_2^2. \quad (7)$$

Because crop organs (such as maize and sorghum leaves) are thin, high-aspect-ratio surfaces, a predicted point cloud can

Model Variant	Sensor Input	Chamfer vs. Target (10^{-4} m^2) Isolated	Row	Δ [95% CI]	Leaf Parameter MAE Isolated	Row	Δ [95% CI]
Maize: Zero-Shot (Headline Model)							
	RGB	56.7	59.7	$+3.0$ [$+2.4$, $+3.6$]	0.085	0.088	$+0.003$ [$+0.003$, $+0.003$]
	Depth	47.2	62.6	$+15.3$ [$+14.8$, $+15.9$]	0.090	0.093	$+0.002$ [$+0.002$, $+0.003$]
	Point Cloud	20.0	33.8	$+13.7$ [$+13.4$, $+14.1$]	0.064	0.081	$+0.018$ [$+0.017$, $+0.018$]
	RGB + D + PC	26.4	29.6	$+3.2$ [$+2.9$, $+3.6$]	0.086	0.093	$+0.007$ [$+0.007$, $+0.008$]
Maize: Row Fine-Tuned							
	RGB	13.7	14.4	$+0.6$ [$+0.4$, $+0.8$]	0.067	0.071	$+0.004$ [$+0.003$, $+0.004$]
	Depth	10.3	9.6	-0.7 [-0.9, -0.5]	0.061	0.064	$+0.003$ [$+0.003$, $+0.004$]
	Point Cloud	21.7	21.9	$+0.2$ [-0.1, $+0.5$]	0.063	0.073	$+0.010$ [$+0.010$, $+0.010$]
	RGB + D + PC	12.6	**4.0**	-8.6 [-8.8, -8.3]	**0.046**	**0.052**	$+0.005$ [$+0.005$, $+0.005$]
Maize: Isolated Control							
	RGB	11.6	56.1	$+44.5$ [$+44.0$, $+45.1$]	0.071	0.095	$+0.024$ [$+0.023$, $+0.024$]
	Depth	4.6	52.8	$+48.2$ [$+47.5$, $+48.9$]	0.059	0.097	$+0.038$ [$+0.037$, $+0.039$]
	Point Cloud	8.2	25.1	$+16.9$ [$+16.6$, $+17.3$]	0.046	0.076	$+0.030$ [$+0.029$, $+0.030$]
	RGB + D + PC	1.8	29.0	$+27.2$ [$+26.9$, $+27.6$]	0.043	0.093	$+0.050$ [$+0.049$, $+0.051$]
Sorghum: Zero-Shot (Headline Model)							
	RGB	163.1	218.5	$+55.4$ [$+52.6$, $+58.4$]	0.129	0.135	$+0.006$ [$+0.006$, $+0.006$]
	Depth	106.2	207.2	$+101.0$ [$+98.0$, $+104.2$]	0.070	0.109	$+0.039$ [$+0.039$, $+0.040$]
	Point Cloud	177.7	231.4	$+53.8$ [$+52.2$, $+55.4$]	0.114	0.127	$+0.013$ [$+0.012$, $+0.013$]
	RGB + D + PC	174.3	230.5	$+56.2$ [$+53.9$, $+58.5$]	0.108	0.114	$+0.006$ [$+0.006$, $+0.006$]
Sorghum: Row Fine-Tuned							
	RGB	34.8	29.8	-5.0 [-5.4, -4.7]	0.094	0.097	$+0.003$ [$+0.003$, $+0.004$]
	Depth	24.4	18.6	-5.8 [-6.1, -5.5]	0.078	0.055	-0.022 [-0.023, -0.022]
	Point Cloud	46.0	70.8	$+24.8$ [$+24.4$, $+25.2$]	0.074	0.082	$+0.007$ [$+0.007$, $+0.007$]
	RGB + D + PC	38.7	**30.7**	-8.0 [-8.5, -7.6]	**0.073**	**0.061**	-0.012 [-0.013, -0.011]
Sorghum: Isolated Control							
	RGB	15.7	133.2	$+117.5$ [$+115.8$, $+119.3$]	0.094	0.114	$+0.020$ [$+0.020$, $+0.021$]
	Depth	6.6	171.9	$+165.3$ [$+162.9$, $+167.8$]	0.042	0.100	$+0.059$ [$+0.058$, $+0.060$]
	Point Cloud	30.9	100.9	$+70.0$ [$+68.8$, $+71.3$]	0.065	0.083	$+0.018$ [$+0.018$, $+0.019$]
	RGB + D + PC	13.1	133.8	$+120.8$ [$+119.5$, $+122.1$]	0.049	0.096	$+0.047$ [$+0.046$, $+0.048$]

Table 7. **Regime III: comparative performance of identical plants in isolation versus crop rows.** Evaluated across 2,250 held-out plants per species at 10 row orientations ($N = 22{,}500$). Chamfer distance is reported in metric units (10^{-4} m^2). Δ indicates the paired within-plant difference (row minus isolated) with 95% bootstrap confidence intervals over plants.

Input Modality	Sorghum (Validation Set) θ MAE ↓ Before	After	Chamfer $\times 10^{-3}$ ↓ Before	After	Maize (Validation Set) θ MAE ↓ Before	After	Chamfer $\times 10^{-3}$ ↓ Before	After
RGB Only	0.058	0.014	2.27	0.88	0.037	0.019	11.3	0.77
Depth Only	0.017	**0.013**	0.90	0.63	0.024	0.019	1.26	0.58
Point Cloud Only	0.040	0.020	5.48	1.39	0.020	0.018	6.43	0.83
θ (Parameters Only)	–	–	22.7	6.83	–	–	80.7	12.8
Two-Stream Input Pairs → Point Cloud Generation (Dedicated Student / Generalist)								
RGB + D	–	–	2.91	0.60 / 0.59	–	–	1.30	0.53 / 0.55
RGB + θ	–	–	0.43	0.40 / 0.39	–	–	0.47	0.44 / 0.41
D + θ	–	–	0.50	0.37 / **0.36**	–	–	0.46	0.42 / **0.40**
Mean Generation Loss	0.371 → 0.259 (−30.2%)				0.290 → 0.164 (−43.5%)			

Table 8. **Any-subset modality reconstruction before and after cross-modal distillation (validation set).** Each row represents evaluation given only the specified source modality to generate remaining streams. Before: zero-shot pre-trained model warm-start; After: distilled generalist student model.

achieve low Chamfer distance simply by clustering points onto bulky central stalks while omitting entire leaf blades. In contrast, downstream linear probes fitted on frozen encoder embeddings directly measure whether latent representations retain the structural parameter information required for phenotyping.

Throughout our experiments, Chamfer distance and linear probes rank model variants differently:

1. **Depth stream addition**: Adding metric depth to RGB photographs reduces Chamfer distance by 28% in sorghum and 25% in maize, but yields negligible change in downstream probe R^2 scores (Tab. 4). Depth ordering provides spatial surface proximity for decoders without expanding high-level trait information in the encoder.
2. **Data scaling**: Increasing pretraining dataset volume from 1,000 to 10,000 plants reduces sorghum Chamfer error

Variant	Vision Budget	θ Vis. Ratio	Chamfer $\times 10^{-3}$	R^2 (Validation Set) Height	Leaf Cnt	Roll	Leaf Len
Sorghum							
PC + RGB + D	117.0	–	**1.85**	**0.977**	**0.980**	**0.781**	**0.309**
$+\theta$, Shared Budget	107.4	58%	2.53	0.962	0.964	0.760	0.249
$+\theta$, Gated 20%	117.0	20%	2.29	0.964	0.967	0.771	0.278
$+\theta$, Gated 60%	117.0	60%	2.40	0.959	0.963	0.761	0.239
$+\theta$, Target Only	117.0	0%	2.49	0.966	0.970	0.702	0.220
Maize				Height	Leaf Cnt	Curl	Stem Rad
PC + RGB + D	117.0	–	1.21	0.633	0.914	0.236	0.760
$+\theta$, Shared Budget	106.6	56%	0.99	0.635	0.908	0.304	0.743
$+\theta$, Gated 20%	117.0	17%	**0.94**	0.704	0.931	**0.464**	0.791
$+\theta$, Gated 60%	117.0	59%	1.04	0.521	0.851	0.303	0.688
$+\theta$, Target Only	117.0	0%	1.16	**0.776**	**0.947**	0.357	**0.804**

Table 9. **Token-budget control ablations.** Four-stream models re-trained with parameter tokens (θ) placed outside the shared Dirichlet budget, maintaining an identical 117.0 visible sensor token allocation across parameter visibility levels (0%, 20%, 60%, and target-only). Maize has 29 parameter tokens against sorghum's 25, so its shares differ slightly; visible shares are averages over masking draws, floored to whole tokens (the maize 20 % arm keeps 5 of 29). Chamfer is each run's final validation value.

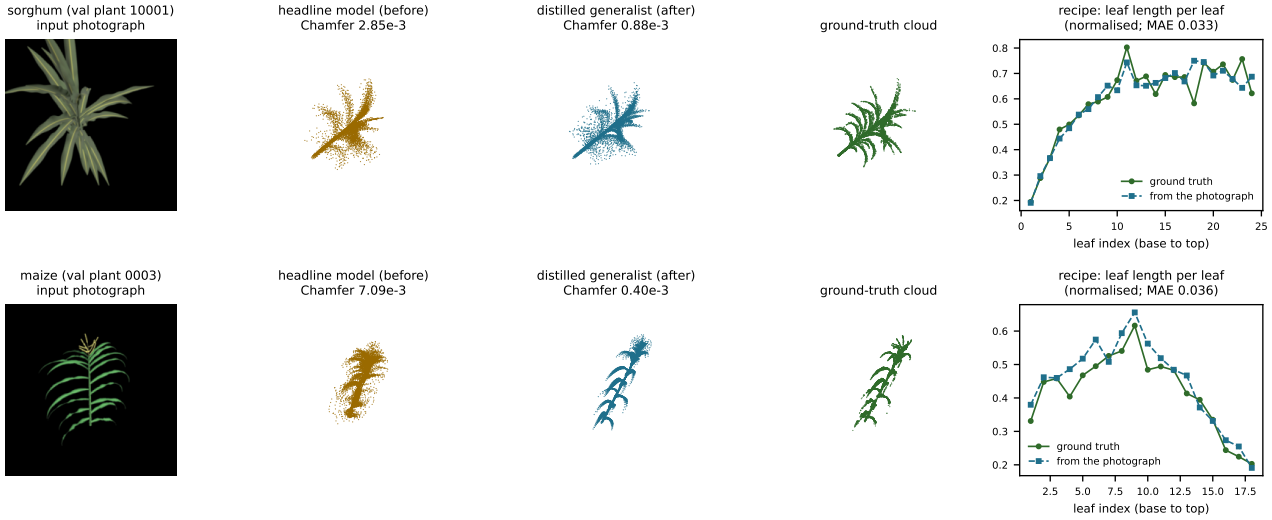


Figure 6. **Single-photograph 3D plant reconstruction on held-out test plants.** Each row shows inference given only a single input photograph (left). The headline model before distillation and the distilled generalist generate 3D point clouds, with Chamfer distance indicated above. The rightmost panel demonstrates parameter recovery by comparing the generalist's recovered normalized leaf lengths with exact ground truth across leaf indices from base to top.

by 21% ($3.13 \rightarrow 2.48 \times 10^{-3}$), whereas sorghum probe scores saturate by 10,000 plants (Tab. 4).

3. **Partial camera visibility**: Restricting point cloud inputs to a single camera perspective increases Chamfer error 2.6× to 6.5×, whereas refitted linear probes of the fused arms lose at most 0.030 in height R^2 (Tab. 6).

These disagreements demonstrate that Chamfer distance alone is an insufficient metric for evaluating slender 3D shape reconstruction. We advocate evaluating structural downstream trait probes alongside Chamfer distance, and incorporating distribution-matching losses (such as Sinkhorn optimal transport divergence, Eq. (4)) during training to force decoders to cover full organ geometry rather than collapsing points onto central stalks.

Partial coverage, camera viewpoint, and row occlusion. Sensor observations in agricultural field campaigns are inherently partial. Our experiments analyze three aspects of partial sensing:

- **Viewpoint elevation dominance**: Across isolated camera perspectives, camera elevation angle dominates single-image reconstruction difficulty, accounting for $\eta^2 = 0.24$–0.31 of per-plant reconstruction variance (Fig. 7). Steep top-down and bottom-up perspectives suffer from severe self-occlusion where leaves stack vertically along the optical axis, whereas side-on views expose full blade

Rows (plants)	Model	Chamfer ↓	Miss@5 mm ↓	Height ↓	Leaves ±1 ↑
Zero-shot, all real plants					
Maize parallel (520)	Headline (no row FT)	75.1 [73.1, 77.2]	0.92	53%	17%
	Isolated control	55.6 [54.2, 57.2]	0.93	43%	4%
	Row fine-tune (FT)	**41.2** [39.7, 42.8]	0.93	**37%**	**47%**
Maize perpend. (520)	Headline (no row FT)	121.2 [116.6, 127.2]	0.96	52%	23%
	Isolated control	137.3 [133.6, 141.9]	0.98	40%	13%
	Row fine-tune (FT)	**80.0** [76.4, 85.2]	0.97	**35%**	**46%**
Sorghum, both (30)	Headline (no row FT)	2,541 [2,205, 2,886]	0.99	–	–
	Isolated control	**975** [876, 1,079]	0.99	–	–
	Row fine-tune (FT)	2,145 [1,910, 2,376]	0.99	–	–
Fine-tuned on 75% of the real plants, scored on the held-out 25%					
Maize, both (130)	Row FT	59.5 [56.0, 63.2]	0.95	36%	45%
	Real-row FT	**19.7** [18.6, 21.0]	**0.87**	**1.6%**	**70%**
Sorghum, both (8)	Row FT	2,113 [1,647, 2,520]	0.99	–	–
	Real-row FT	**79.4** [68.3, 92.6]	**0.92**	–	–

Table 10. **Real plants in crop rows (E10).** All three streams in. Chamfer of the center plant in 10^{-4} m^2 with a 95% bootstrap interval over plants; Miss@5 mm is the share of the true center plant's points farther than 5 mm from the prediction; height error is the median relative error; Leaves ±1 is the share of stops whose predicted leaf count is within one of the true count. Sorghum has no comparable height or leaf-count truth (–). Real-row FT is the row fine-tune further trained on the other 75% of the real plants. On synthetic test rows the row fine-tune reaches a Chamfer of 4.0 (maize) and 30.7 (sorghum).

	Maize (130 plants)		Sorghum (8 plants)	
Input	Row FT	Real-row FT	Row FT	Real-row FT
Image + depth + cloud	59.5	**19.7**	2,113	**79.4**
Image only	95.4	48.7	1,033	106.2
Depth only	76.8	43.6	834	89.0
Cloud only	75.0	47.4	3,773	311.8

		Height error		Chamfer	
Maize plant height	Plants	Row FT	Real-row FT	Row FT	Real-row FT
≤1.33 m (training range)	22	15%	**2.0%**	47.0	**16.3**
1.33–1.6 m	45	30%	**1.2%**	57.7	**18.8**
1.6–2.0 m	51	41%	**1.4%**	62.7	**20.4**
>2.0 m	12	51%	**5.9%**	75.4	**26.7**

Table 11. **Real-row fine-tune in detail (E10, held-out plants).** Top: Chamfer of the center plant (10^{-4} m^2) for each input, before (row fine-tune) and after fine-tuning on 75% of the real plants. Bottom: maize, all three streams in, split by the plant's true height; height error is the median relative error. The recipe's height range ends at 2.0 m.

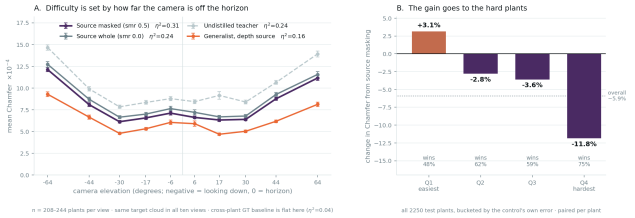


Figure 7. **Viewpoint elevation determines reconstruction difficulty.** (A) Photograph-to-cloud Chamfer distance of test plants plotted by camera elevation angle (208–244 plants per elevation view). Pretrained and distilled single-sensor models follow a symmetric U-shaped error curve centered on horizontal side views; camera elevation accounts for $\eta^2 = 0.24$–0.31 of per-plant reconstruction variance. (B) Performance gain from masking 50% of source tokens during distillation, grouped by error quartile of the control model, demonstrating concentrated improvement on challenging plant geometries.

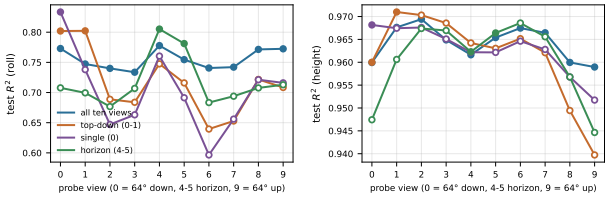


Figure 8. **Effect of training view distribution on downstream phenotype probing (sorghum test).** Probes are fitted and evaluated on individual views k for model variants trained on all ten views versus restricted bands of the elevation ladder. Filled markers indicate training views. Restricted training view bands improve probe performance specifically within those elevations but degrade generalization on unobserved view angles.

silhouettes.

- **Covariate shift under partial visibility**: While phenotypic trait information persists in encoder representations under partial surface coverage (refitted 1-camera height $R^2 = 0.960$), directly transferring linear probes trained on 100% complete clouds to partial inputs results in performance failure ($R^2 \to -0.26$, Fig. 9). Partial visibility induces a systematic covariate shift in latent feature space. Consequently, downstream phenotyping readouts deployed in the field must be refit or calibrated on sensor-matched partial coverage inputs.
- **Implicit row occlusion reasoning**: In planted rows, neighboring foliage accounts for up to two-thirds of input point clouds and pixel areas. Standard models trained on isolated plants suffer catastrophic failure under row occlusion because decoders attempt to incorporate neighboring leaves into target plant reconstructions (Tab. 7). Fine-tuning decoders to predict target main plant geometry behind occluding neighbors removes 92% of neighbor-induced occlusion error in maize and 85% in sorghum, forcing the network to perform implicit occlusion reasoning and segment target organs from inter-plant contact foliage.

Importantly, procedural parameter outputs ($\hat{\theta}$) demonstrate superior robustness to partial visibility compared to raw point cloud completions. Because the procedural generator $G(\hat{\theta})$ acts as an explicit biological prior, re-rendered procedural meshes generate complete 3D geometry even when input sensor observations capture only a fraction of the plant surface.

Capabilities and boundaries of cross-modal distillation. Cross-modal teacher–student distillation (Sec. 3.4) converts

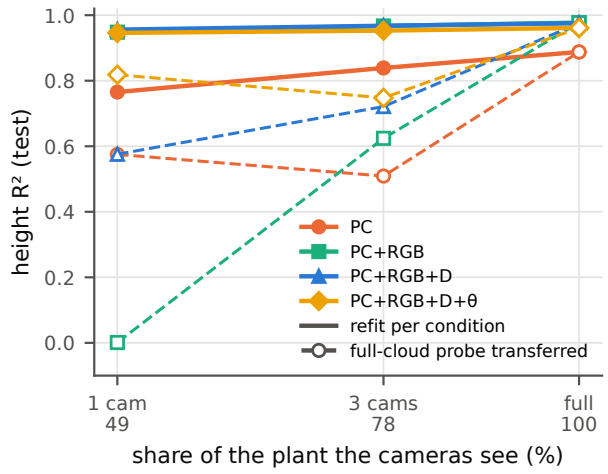


Figure 9. **Phenotypic information persistence under partial visibility.** Plant height prediction performance (R^2, test set) plotted against the visible percentage of the plant under 1-camera, 3-camera, and full coverage. Solid lines represent probes refit per visibility condition; dashed lines show direct transfer of the full-cloud readout probe.

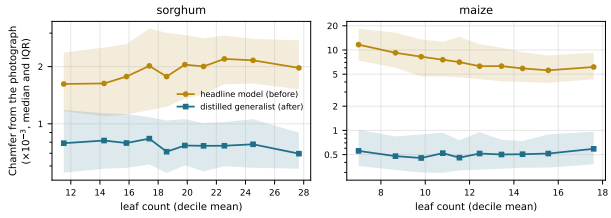


Figure 10. **Error against plant complexity (test, 2,250 plants per species).** Chamfer distance of point clouds generated from single photographs across leaf-count deciles for the headline model before distillation and the distilled generalist.

a multimodal pretraining artifact into a fast, deployable single-sensor inference model. Distilling a full-modality teacher (RGB + D + PC + θ) into single-sensor student networks reduces overall cross-modal generation loss by 30.2% in sorghum and 43.5% in maize (Tab. 8). Distillation gains are concentrated on the weakest input modalities (photograph-to-cloud Chamfer error falls by 61%) and the most challenging plant geometries (Fig. 7B). Furthermore, supervising student models across all four decoder targets simultaneously outperforms a specialized parameter-only regression student (recipe MAE from the point cloud 0.019 against 0.036 and 0.038 for specialists trained at 10^{-4} and 3×10^{-5}, from the same warm start at 0.040), consistent with auxiliary visual reconstruction tasks acting as regularizing gradients for parameter regression.

However, distillation exhibits clear boundary limits:

1. **Viewpoint robustness**: Distillation lowers absolute reconstruction error across all camera views but does not flatten the U-shaped elevation difficulty curve (Fig. 7A). Viewpoint robustness cannot be acquired purely through distillation; it requires balanced elevation sampling during pretraining data capture (Fig. 8).
2. **Unobservable parameter instability**: Parameterizations submitted to inverse regression must be restricted to traits that field sensors can physically resolve. In early experimental iterations containing unobservable leaf waviness parameters, distillation caused these unconstrained parameter heads to develop severe view-dependent instability without improving trait skill (skill ≈ 0). Restricting procedural parameter representations to observable physiological attributes (insertion heights, blade lengths, widths, angles, and rolls) ensures stable cross-modal distillation.

Limitations and future directions. We acknowledge several limitations in our current study:

- **Single-seed ablation budgets**: Controlled ablation experiments (Tabs. 4 and 9) were executed under a standardized compute budget of 197,400 optimizer steps using a single seed per arm. Run-to-run baseline variance estimated from near-duplicate runs is $\approx 2\%$ in Chamfer distance and ≤ 0.011 in probe R^2; the non-monotone maize scaling differences, though significant under the paired bootstrap, could still reflect run-to-run variance.
- **Procedural generator expressivity boundaries**: An inherent limitation of inverse procedural modeling is that recovered representations are bounded by the expressivity of the underlying generator G. If a physical plant exhibits morphological deformations unrepresented in G (such as severe disease lesions, structural breakage, or tassel droop, which the generator renders without learnable variation), the framework fits the nearest valid procedural parameter configuration. Expanding procedural generators to support broader developmental malformations represents an important avenue for future research.
- **Synthetic renderer regularities**: Synthetic rendering pipelines can introduce unintended canonical cues (such as fixed leaf plane azimuth alignments) that pose-aware baselines may exploit as shortcuts. Rigorous evaluation on real-world field captures (Sec. 4.8) remains essential to verify true sim-to-real domain transfer.

By releasing FIELDSCAN (comprising 30,000 synthetic plants across maize and sorghum with paired multimodal captures, physically simulated row occlusion, and exact procedural parameter ground truth), we aim to provide the 3D vision and digital agriculture communities with a benchmark for evaluating multimodal completion and inverse procedural modeling on slender, heavily occluded geometry.

E. Detailed Literature Review

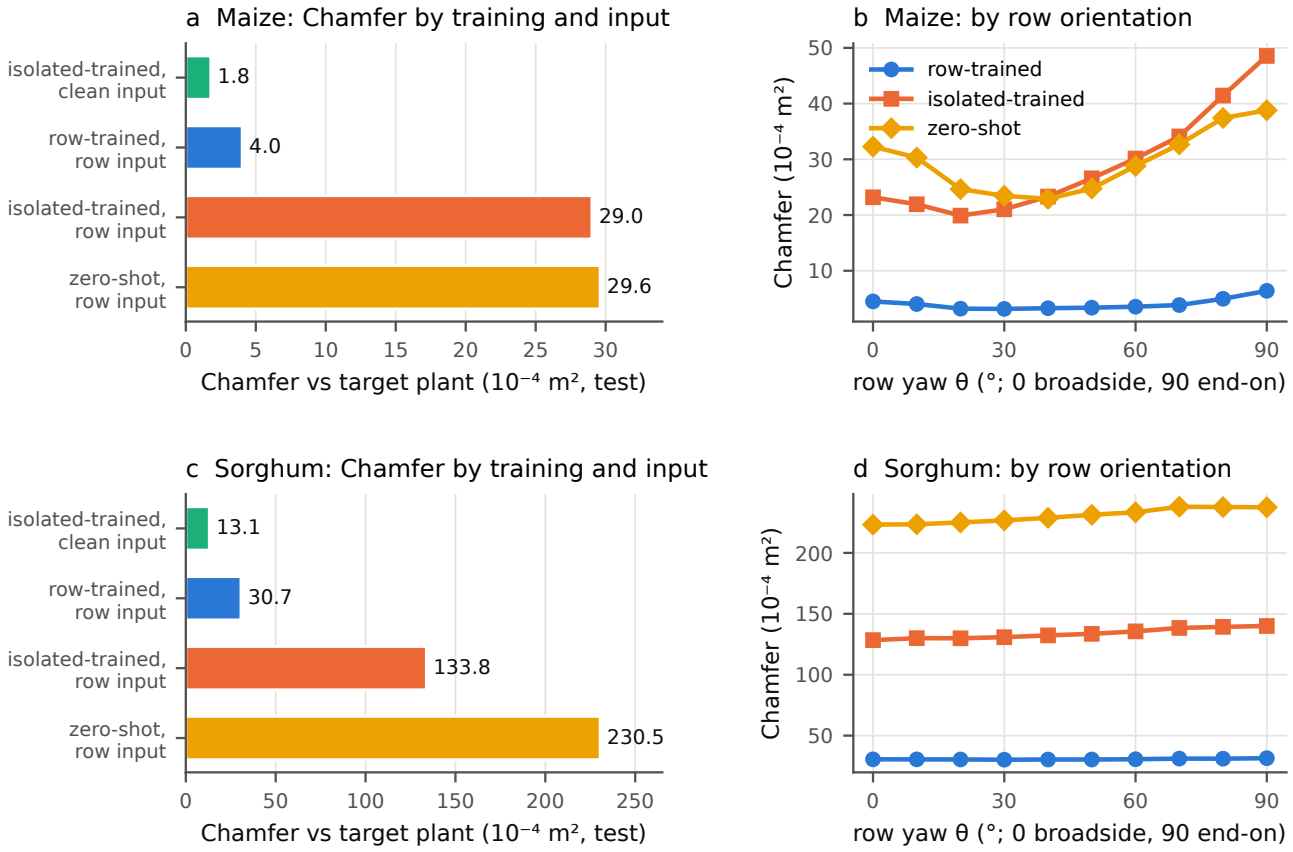


Figure 11. **Quantitative row occlusion performance on maize (a, b) and sorghum (c, d) test rows.** (a, c) Target plant Chamfer distance comparing isolated-trained models on clean versus row inputs, row-trained models on row inputs, and zero-shot baselines. (b, d) Reconstruction error plotted across row orientation yaw angles (θ). Fine-tuning on row scenes eliminates the vast majority of neighbor-induced occlusion error in both species.

This appendix expands the compressed related-work discussion of Section 2 across six core bodies of literature on which CROPCODE builds and from which it departs.

E.1. Procedural plant models and inverse procedural modeling

Procedural modeling of plant architecture originated with Lindenmayer systems (L-systems) [35] and matured into parametric generators whose parameters encode physiologically meaningful organ-level attributes [19, 20]. Inverting such models, recovering generative parameters from visual observations, has a long history for woody trees: Stava et al. fit tree generator parameters via stochastic search [43], Talton et al. formulated inverse procedural modeling as Markov Chain Monte Carlo (MCMC) inference [44], and Guo et al. inferred structural L-systems from 3D point clouds [17], with deep generative variants following [28].

These methods optimize parameters on a per-instance basis. None performs amortized, single-forward-pass inversion of a procedural plant generator from noisy, field-degraded sensor data. The closest feed-forward precedents lie outside agriculture: shape-program induction [45], diffusion-based inverse procedural content generation from single photographs [53], text-conditioned CAD generation [24], and cadrille, which maps images, point clouds, or text to executable CAD programs [26]. However, these systems operate on rigid, clean, isolated man-made objects. None addresses deformable slender organs, heavy inter-plant row occlusion, or real-world sensor noise, and none investigates whether continuous program parameters should enter the transformer encoder during multimodal pretraining.

E.2. Learned parametric recovery of crop plants

Recent systems attempt to recover parametric plant models from 3D scans. Demeter learns a joint morphology model (topology, articulation, and PCA surface variations) from soybean point clouds [8], while Ghrer et al. infer grammar-style parameterized crop representations from 3D

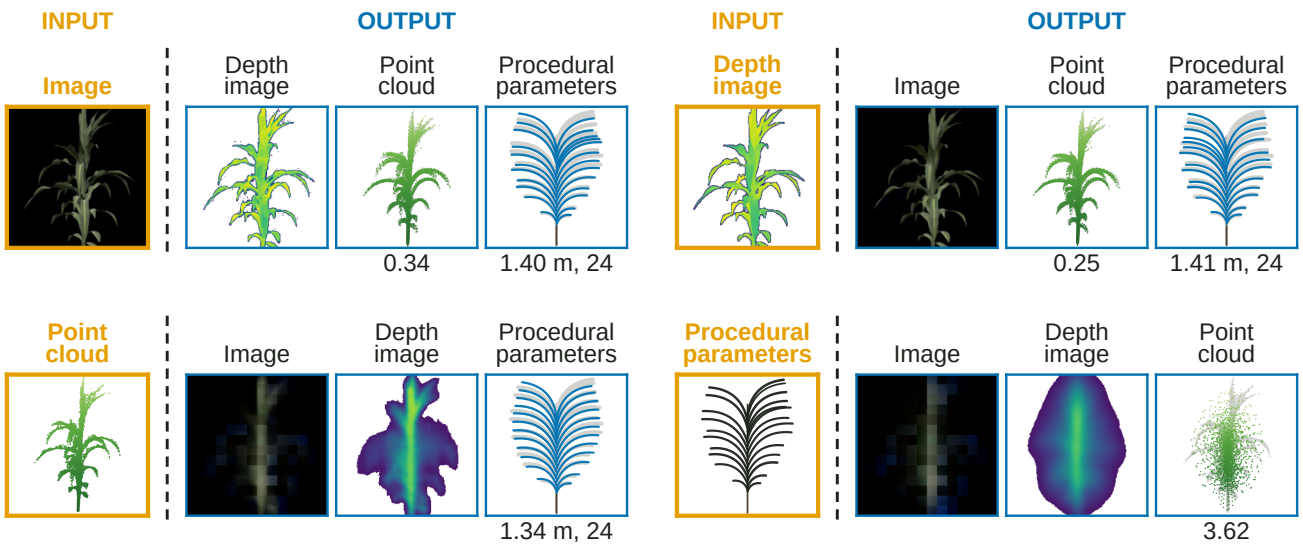


Figure 12. **Any-to-all, sorghum: each stream alone in, the other three out.** The distilled generalist on one held-out test plant. Each block gives it one stream (orange, left of the dashed line) and shows the three streams it generates (blue). Generated point clouds and procedural parameters are drawn over the ground truth in grey, the parameters as a stem-and-leaves schematic. Under each generated point cloud: its Chamfer distance (10^{-3}); under each generated parameter set: height and leaf count (truth 1.43 m, 24 leaves). The plant is selected for clarity among the test plants with the lowest image-to-cloud Chamfer, seen within 25° of the horizon. The parameters alone give only a coarse plant (Chamfer 3.62 here, 6.83 on average in Table 8).

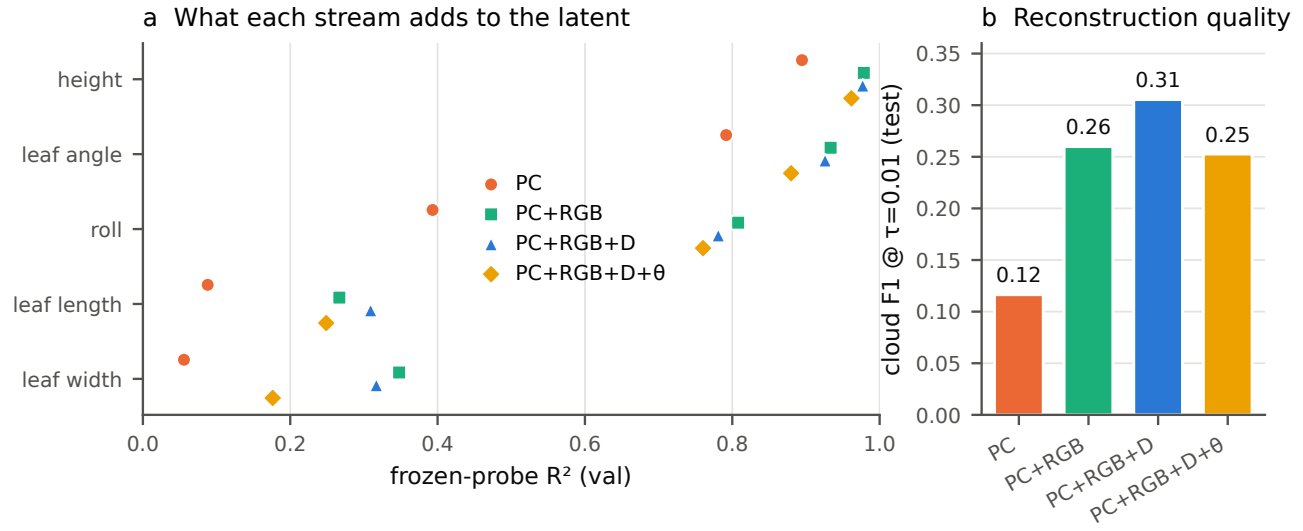


Figure 13. **Modality contributions to latent representation and geometry reconstruction (sorghum).** (a) Frozen-probe R^2 across phenotypic traits. (b) Reconstruction F_1 score at threshold $\tau = 0.01$ (≈ 1 cm).

laser scans [14]. Both systems assume a single, clean, highly complete point cloud as input, and neither addresses missing sensor modalities or inter-plant row occlusion.

CropCraft recovers procedural crop structure by applying Bayesian optimization over procedural parameters matched against multi-view NeRF depth maps [52]; however, per-scene optimization requires substantial computation per plant. A staged annotation pipeline processes terrestrial laser scans through vision-language model prompting and fits procedural NURBS descriptors via differentiable refinement [9, 18, 19], requiring $\sim$ 10 minutes of computation per isolated plant. Tree-D Fusion reconstructs simulation-

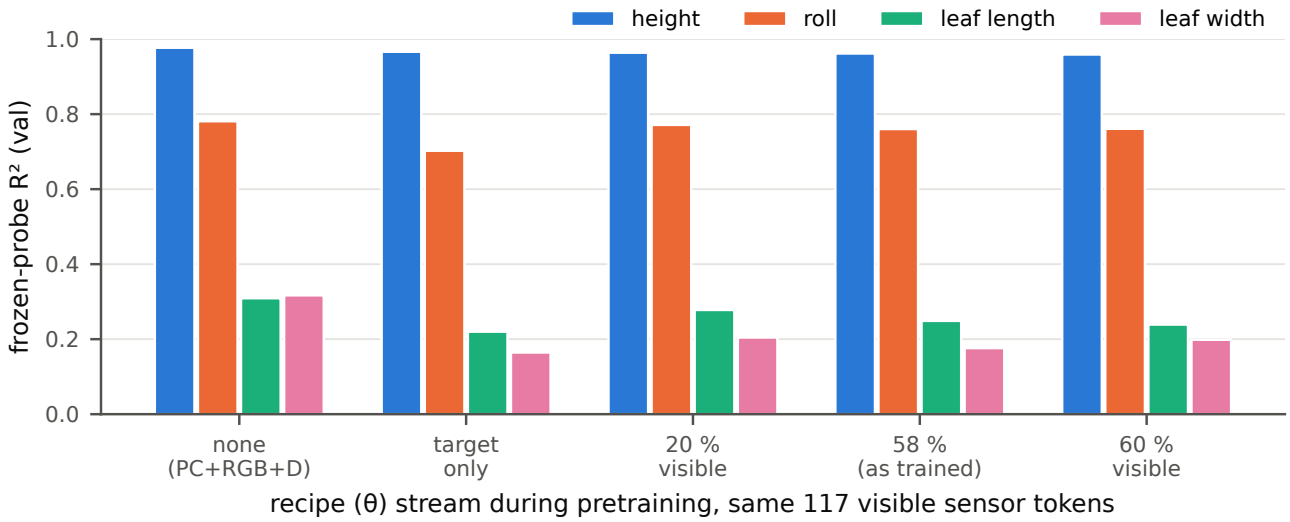


Figure 14. **Influence of procedural parameter stream visibility during pretraining.** Linear probe R^2 performance across parameter visibility levels under matched sensor token budgets.

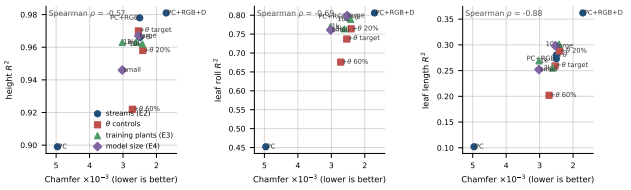


Figure 15. **Divergence between Chamfer distance and frozen-latent probe metrics.** Validation Chamfer distance plotted against height probe R^2 across twelve model variants.

ready urban trees from street-level imagery via diffusion priors [29], but its envelope-then-branch pipeline targets woody tree silhouettes rather than thin, deformable agricultural crop blades.

E.3. Point cloud completion and slender geometry

Learned point cloud completion has advanced from folding-based decoders [51] to geometry-aware transformers [49, 50] and extrapolation–interpolation architectures [5]. Standard benchmarks (e.g., PCN, ShapeNet-55/34) consist of compact, largely convex rigid objects and contain no slender-under-occlusion categories. Standard completion algorithms assume missing geometry represents self-occluded object surfaces without severe inter-object clutter.

Plant-specific completion has recently emerged, including amodal segmentation and geometry completion for occluded crops [27]. However, these methods output unstructured point clouds rather than editable, biologically interpretable procedural parameters. Curve-aware and thin-structure reconstruction methods [13, 38] address slender geometry, a regime in which standard Chamfer distance fails due to structural insensitivity to thin surfaces [42, 47]. This limitation has motivated density-aware and information-theoretic distance variants [30, 47]. In this work, we systematically compare surface geometric metrics against downstream linear trait probes on frozen encoder latents, demonstrating that the two metrics rank architectural choices differently on slender plant geometry.

E.4. 3D plant phenotyping

Organ-level 3D phenotyping datasets, including Pheno4D's annotated maize and tomato temporal sequences [41], Crops3D [54], and field-scale maize scans carrying procedural reference fits [25], support trait extraction using deep point cloud architectures [36, 37]. Furthermore, field-based radiance-field capture is increasingly viable [2]. However, no existing benchmark combines real-field captures, slender plant geometry, physically simulated inter-plant row occlusion, and exact procedural ground-truth parameter targets across multiple crop species. FIELDSCAN is designed to fill this gap.

E.5. Multimodal masked pretraining and distillation

Masked autoencoding [21] has been extended to multimodal domains: MultiMAE over RGB, depth, and semantic segmentation [3]; 4M and 4M-21 over tokenized visual, structural, and textual modalities [4, 31]; Point-MAE over 3D point clouds [33]; and EmbodiedMAE over RGB, depth, and point clouds for robotic manipulation [10].

Maintaining robustness under missing input modalities

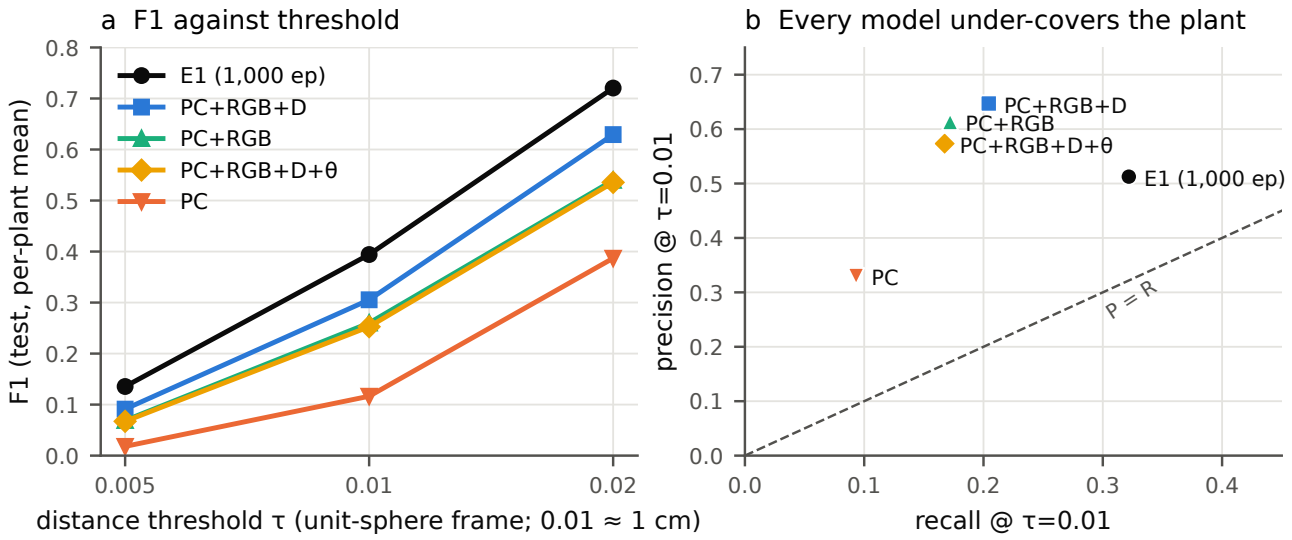


Figure 16. **Point cloud completion precision, recall, and F_1 evaluation (sorghum test).** (a) F_1 score against distance threshold τ. (b) Precision vs. recall at $\tau = 0.01$.

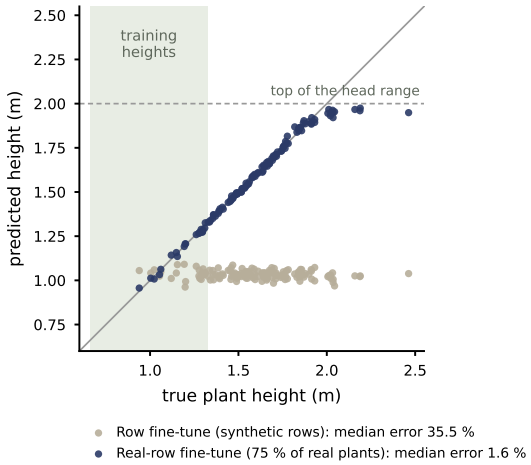


Figure 17. Predicted against true plant height on the 130 held-out real maize plants (median over each plant's 36 camera stops, all three streams in). Trained on synthetic rows only, the model writes about 1.03 m for every plant; after fine-tuning on the other 75% of the real plants it follows the true height up to the 2.0 m end of the recipe's range. The shaded band is the training height range.

is an active area of research [40]. Cross-modal feature distillation from multi-sensor teacher models to single-sensor student models is established for discriminative tasks [22]. We extend cross-modal distillation to a *generative* setting for structured parameter regression, encoding continuous procedural parameters as numerical tokens via xVal-style embedding [15]. Our finding that exposing ground-truth parameter tokens to the encoder degrades visual representation quality connects to modality competition [34] and gradient interference [7, 48], establishing a concrete design rule for privileged generative modalities in multimodal pretraining.

E.6. Synthetic-data pretraining and sim-to-real transfer

Pretraining deep representations on synthetic data is established across domains ranging from fractal geometry [23] to procedural 3D environments [16, 39], with domain randomization closing the sim-to-real gap [46]. The utility of synthetic data is known to depend heavily on downstream task alignment [11], and model self-consumption presents documented stability risks [1].

Our setting represents an ideal use-case for synthetic pretraining: the deployment target is the exact parameter space of the procedural generator, the generator spans the continuous morphological variation of target crop species, and the framework is validated against high-density real-world field captures [25].

F. Detailed Ablation Studies

All reported ablation experiments are evaluated on the complete validation and test sets of both species across standardized 197,400 optimizer step budgets at global batch size 32. Representation quality is evaluated using ridge regression probes fit on frozen `[CLS]` tokens (one deterministic perspective per plant) with paired bootstrap confidence intervals (2,000 resamples over plants). A performance difference is classified as significant only when its 95% confidence interval excludes zero on both validation and test splits simultaneously.

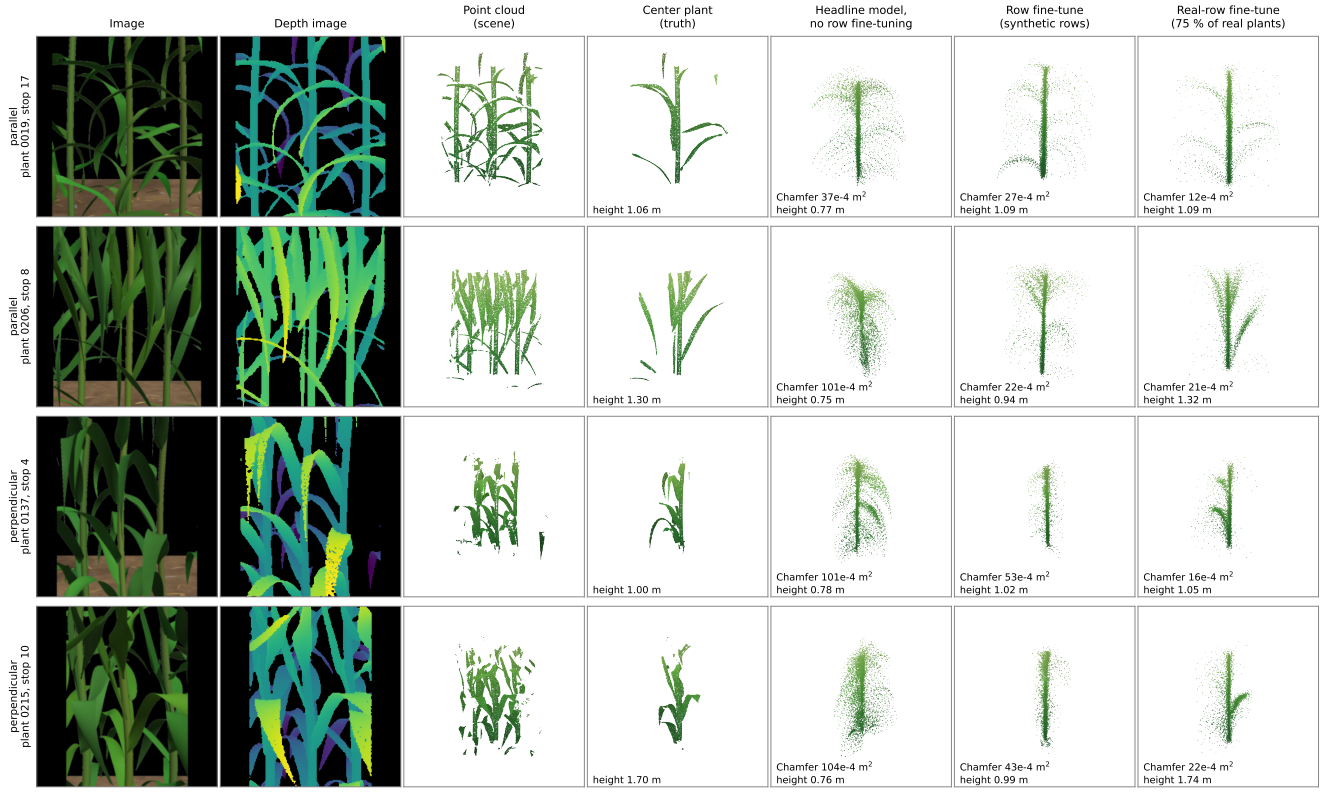


Figure 18. Real maize rows (E10), held-out test plants. Each row of panels is one camera stop on a crop row of a real maize plant that the real-row fine-tune never saw, in the parallel and perpendicular layouts, picked at its median Chamfer for the layout. From left: the image, depth image and scene point cloud the model reads, the center plant as the camera sees it, and the center plant written by the headline model without row fine-tuning, by the row fine-tune trained on synthetic rows, and by the real-row fine-tune trained on 75% of the real plants, each with its Chamfer and predicted height. Row fine-tuning removes the neighbors; fine-tuning on real plants recovers the leaf layout and the height.

F.1. Training view distribution

We pretrain models under restricted camera elevation bands: top-down views ($i \in \{0, 1\}$), horizontal views ($i \in \{4, 5\}$), or a single top-down view ($i = 0$), benchmarked against pretraining across all ten views. Probes are evaluated across held-out views k (Fig. 8). Restricted models achieve superior performance on their specific training elevations (e.g., $+0.061$ roll R^2 on view 0 for the single-view model), but experience performance drops on unobserved elevations (falling by -0.066 to -0.144 on distant views). Models trained on the complete elevation ladder maintain stable performance across all perspective angles ($R^2 = 0.734$–0.778). Maize gives the same answer without a per-view probe: its cameras sit at random elevations, so the probe camera is a random one in every arm. All three restricted arms lose to the all-view arm on every target (validation R^2 for height 0.45, 0.49 and 0.52 against 0.64 for the top-down, horizontal and single-camera arms; leaf angle 0.82, 0.80 and 0.89 against 0.92; stem radius 0.57, 0.57 and 0.72 against 0.75). The single-camera arm trains on one tenth of the images but at the probe's elevation distribution and loses least; the two band arms train on twice as many images inside a band and lose most, so covering the evaluation elevations matters more than the number of images. Paired bootstraps over plants confirm it: against the all-view arm the horizontal arm loses all 11 probe targets and the top-down and single-camera arms lose 10 (height -0.15, -0.19 and -0.12 on validation, 95 % intervals of about ± 0.02); no restricted arm wins a target. Reconstruction gives the same reading through the point-cloud F_1 (protocol of Fig. 16, $\tau = 0.01$, test split). In sorghum the score is taken on view 0, a top-down camera: the two arms trained on that elevation reach 0.27 and 0.28 against 0.25 for the all-view arm, while the horizontal arm, which never saw such a camera, falls to 0.03. In maize the score is taken on a random camera, and every restricted arm falls below the all-view arm's 0.31 (horizontal 0.22, top-down 0.19, single camera 0.17).

F.2. Modality value-add ablations (E2)

Eight pretraining arms per species were evaluated under

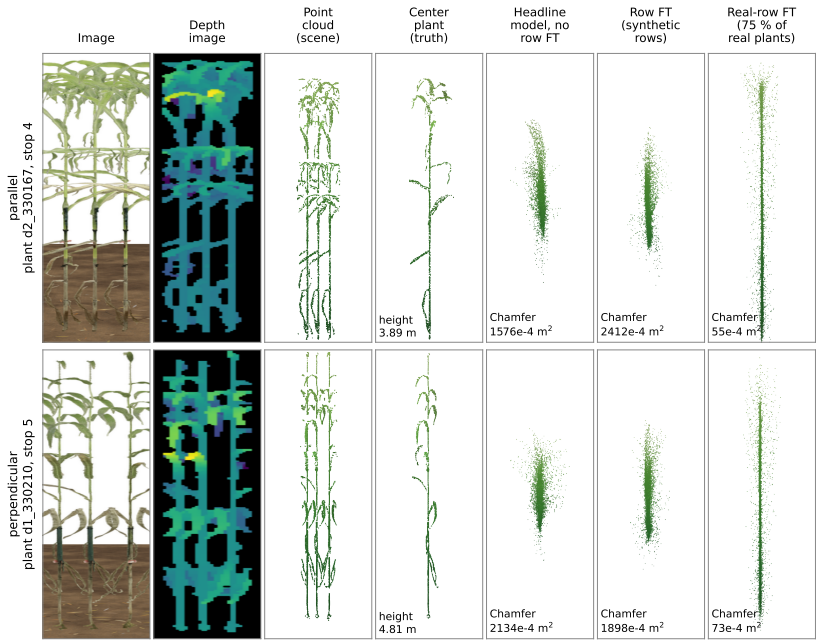


Figure 19. Real sorghum rows (E10), held-out test plants. Columns as in Fig. 18; the image is the camera photograph, a tall strip framed to the whole plant. The plants are 2.2–5.3 m tall; the stems of the training plants reach at most 1.74 m. Without real data every model writes a short, dense plant; the real-row fine-tune, trained on 19 real plants, recovers the stalk over the full plant height but few of the leaves.

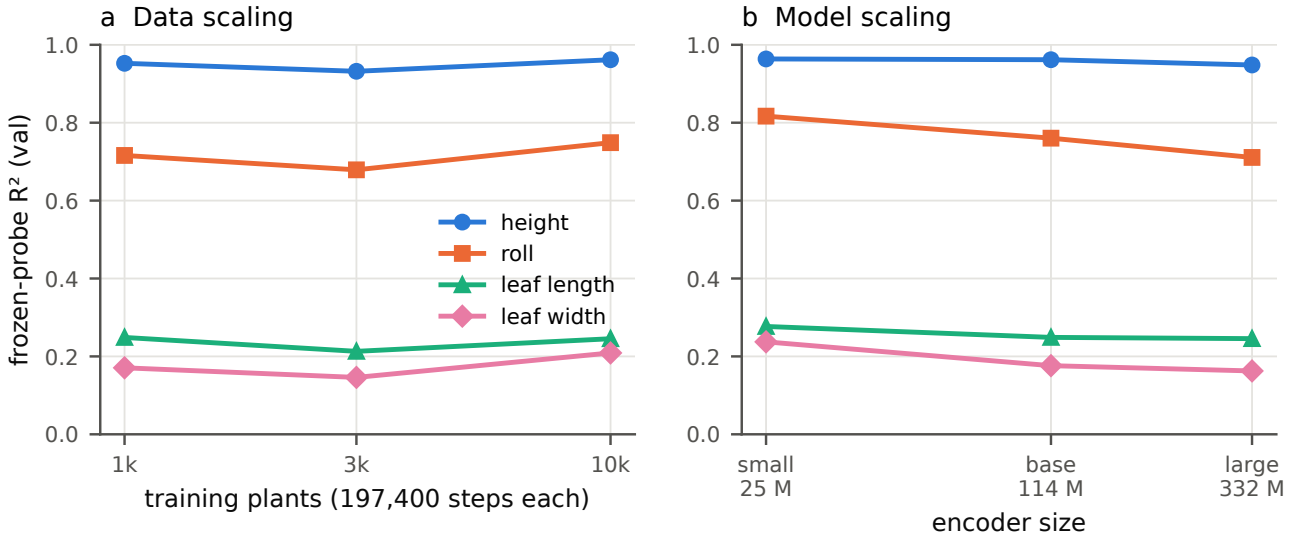


Figure 20. **Dataset and model capacity scaling at equal step compute (sorghum).** Linear probe validation R^2 plotted against dataset size (a) and encoder parameter capacity (b).

identical optimizer step budgets, differing strictly in active token streams (Tab. 12).

F.3. Token-budget control arms

In default four-stream pretraining, sensor streams share a total visible budget of 122–123 tokens with parameter tokens (yielding 107.4 visible sensor tokens vs. 117.0 in three-stream models, with parameter tokens $\sim 58\%$ visible). Control arms decouple parameter tokens from the sensor token allocation, restoring a full 117.0 visible sensor token allocation across three parameter visibility levels: 20%, 60%, and 0% (target-only).

Experimental Variant	Sorghum Chamfer ($\times 10^{-3}$)	Sorghum Height R^2	Maize Chamfer ($\times 10^{-3}$)	Maize Height R^2
Point Cloud (PC)	5.00	0.895	2.63	0.171
PC + Depth	2.45	0.976	2.55	0.453
PC + RGB	2.57	0.979	1.61	0.620
PC + RGB + Depth	1.85	0.977	1.21	0.633
PC + Parameters (θ)	4.84	0.860	2.49	0.256
PC + Depth + θ	4.08	0.907	2.54	0.382
PC + RGB + θ	3.00	0.962	1.21	0.509
Full Four-Stream Model (PC + RGB + Depth + θ)	2.53	0.962	0.99	0.635

Table 12. **Extended modality value-add ablations (E2, validation set).** In sorghum, including parameter tokens in the encoder increases Chamfer error across sensor sets (except PC alone) and degrades linear probe R^2. In maize, parameter inclusion lowers surface Chamfer distance while its impact on probing depends on the sensor set.

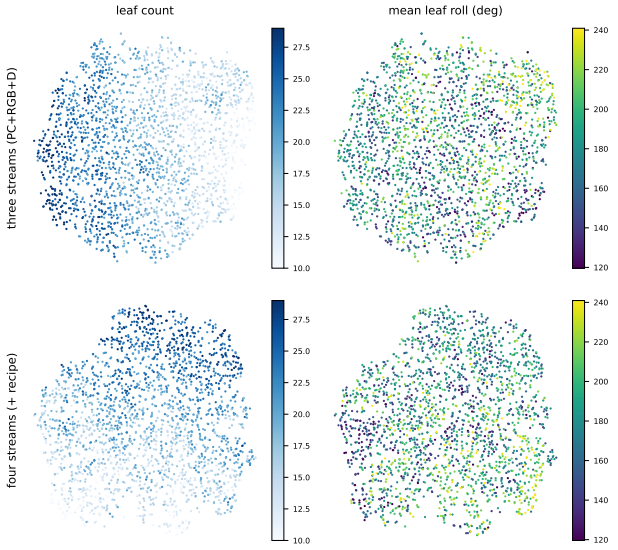


Figure 21. **Latent space visualization (sorghum validation set).** t-SNE projections of frozen encoder `[CLS]` tokens for 3-stream (PC + RGB + D) and 4-stream (+θ) models, colored by plant leaf count and mean leaf roll.

In sorghum, restoring sensor token budgets does not recover three-stream probing performance: height R^2 remains 0.959–0.966 (vs. 0.977), and leaf roll R^2 remains 0.702–0.771 (vs. 0.781). In maize, 20% parameter visibility and target-only parameter supervision outperform both the shared-budget four-stream model and three-stream PC + RGB + D across all eleven traits on both splits.

F.4. Parameters as a decoder reconstruction target only

Restricting procedural parameters strictly to a decoder reconstruction target ($\rho_\theta = 1.0$, never encoded as input) still degrades sensor representations in sorghum relative to omitting the stream: height R^2 drops by -0.011, leaf roll by -0.07, mean leaf length by -0.10, and leaf width by -0.15 on validation. The learned latent space becomes heavily dominated by global plant scale: the first principal component accounts for 44% of total latent variance and correlates strongly with leaf count ($|r| = 0.95$), while the participation ratio drops to 4.3 (vs. 6.7 for PC + RGB + D). Conversely, in maize, target-only parameter supervision ($\rho_\theta = 1.0$) yields the top-performing four-stream model, outperforming PC + RGB + D across all eleven traits.

F.5. Data volume and model capacity scaling at fixed compute

Table 13 evaluates dataset volume scaling (1k, 3k, 10k, 10.5k plants) and model parameter capacity (ViT-Small 25 M, ViT-Base 114 M, ViT-Large 332 M) under matched 197,400 optimizer step budgets.

F.6. Distillation dynamics per source modality

Distillation from the full-modality teacher reduces overall cross-modal generation loss by 30.2% in sorghum and 43.5% in maize (evaluating epoch-0 warm-start initialization against epoch-100 distilled students). Gains are concentrated on the weakest input modalities: point-cloud-to-cloud Chamfer error in sorghum drops from 5.48×10^{-3} to 1.39×10^{-3}, parameter-only-to-cloud Chamfer drops from 22.7×10^{-3} to 6.83×10^{-3}, and photograph-to-cloud Chamfer in maize falls from 11.3×10^{-3} to 0.77×10^{-3}. Parameter regression MAE from point clouds alone is halved in sorghum ($0.040 \rightarrow 0.020$). The parameter-only specialist was retrained on the final data from the same headline warm start with the first-release recipe (point-cloud source, half of its tokens hidden, recipe target only, 100 epochs), once at the generalist's learning rate of 10^{-4} and once at the first-release rate of 3×10^{-5}. Scored on the full validation split with the point cloud fully visible, the specialists reach a recipe MAE of 0.036 and 0.038 (warm start 0.040), the generalist 0.019: the student that also generates the photograph, the depth frame and the cloud recovers the recipe better than the

Dataset Scale / Model Capacity	Sorghum Chamfer ($\times 10^{-3}$)	Sorghum Height R^2	Maize Chamfer ($\times 10^{-3}$)	Maize Height R^2
1,000 Plants	3.13	0.952	3.54	0.393
3,000 Plants	2.86	0.932	1.46	0.524
10,000 Plants	2.48	0.962	0.87	0.427
10,500 Plants (Full Dataset)	2.53	0.962	0.99	0.635
ViT-Small (25 M parameters)	2.86	0.964	1.22	0.563
ViT-Base (114 M parameters)	2.53	0.962	0.99	0.635
ViT-Large (332 M parameters)	2.68	0.948	1.06	0.580

Table 13. **Extended data and model scaling metrics (validation set).** Four-stream model evaluated across dataset sizes and encoder capacities at equal 197,400 optimizer step budgets. Scaling to ViT-Large helps neither species at this budget; ViT-Small wins sorghum roll and leaf length.

student trained on the recipe alone, at either learning rate.

F.7. Label-free view invariance across camera perspectives

Because ten camera perspectives per plant share identical ground-truth procedural parameters, prediction variance across perspectives provides a label-free metric of view invariance. Teacher–student distillation reduces both absolute parameter error and prediction spread across camera views by $\sim 50\%$ on fields with skill (insertion height error reduced by 59%, branching angle error reduced by 60%). Zero-skill fields instead become more view-dependent under distillation, as the head starts to track image noise; the final generator drops those fields.

F.8. Frozen-encoder baselines and pose cues

On sorghum, three-stream CROPCODE (PC + RGB + D) outperforms all published self-supervised visual encoders (DINOv2 [32], MultiMAE [3], Point-MAE [33], and EmbodiedMAE-B [10]) across all traits. Paired over plants against EmbodiedMAE-B, CROPCODE wins 8 of 9 targets (height R^2 $+0.005$, roll $+0.04$, leaf length $+0.08$, leaf width $+0.09$). On maize, four-stream CROPCODE outperforms EmbodiedMAE-B on 10 of 11 traits.

Evaluating Point-MAE with full 3D camera pose matrices yields inflated performance on synthetic maize due to a renderer artifact: synthetic maize scenes align leaf planes along a canonical world plane, enabling full pose matrices to leak plant azimuth orientation. Evaluating Point-MAE with gravity-aligned camera poses (realistic field calibration) reduces its height R^2 from 0.720 to 0.588, while pose-free CROPCODE wins 5 of 9 decided traits.

F.9. Masked reconstruction against published encoders

Table 3 compares frozen features. Table 14 compares what the encoders that have a decoder reconstruct when part of the input is hidden, on the same 2,250 validation plants per species and view 00. Image block: the photograph and the depth image are given with 75% of each image's 16-pixel patches hidden (MultiMAE's pretraining budget) and the point cloud hidden; MultiMAE-B is compared with the CROPCODE encoders that read images. Both models predict per-patch standardized color, so both are de-standardized with the true patch mean and variance, the usual MAE convention, and the photograph is scored by PSNR on the plant pixels of the hidden patches. Each model's normalized depth is mapped back with the two per-image numbers its own normalization used and scored in millimeters on the same pixels. Cloud block: the point cloud alone is given with 60% of its point groups hidden (Point-MAE's pretraining ratio). Point-MAE copies the visible 40% of the points into its output and is given the centers of the hidden groups, as in its pretraining; CROPCODE receives neither and generates the whole cloud. Prediction and truth are each subsampled to 2,048 points and compared in the unit-sphere frame by Chamfer and by precision, recall and F1 at $\tau = 0.01$ and 0.02.

The three-stream CROPCODE encoder reconstructs hidden photograph patches better than MultiMAE in both species (28.0 against 24.7 dB in sorghum, 19.7 against 14.4 dB in maize); the headline model is below MultiMAE in sorghum (23.4 dB) and above it in maize (18.1 dB). Depth is CROPCODE's weak stream under this protocol. MultiMAE is better in sorghum (RMSE 280 against 400–483 mm), and the three-stream encoder only matches it in maize (216 against 215 mm). A predictor that fills every hidden plant pixel with the mean depth of the visible plant pixels reaches 336 mm in sorghum and 236 mm in maize (300 plants), so in sorghum the CROPCODE depth reconstruction is worse than this trivial baseline. Point-MAE reconstructs the cloud far more closely (F1 at $\tau = 0.01$: 0.26 against at most 0.13 in sorghum, 0.52 against at most 0.16 in maize), which its copied visible points and given group centers explain in part. Among the CROPCODE encoders, the headline model reconstructs the cloud best in both species. Its precision at

	Sorghum			Maize		
Image block	PSNR ↑	Depth RMSE ↓	Depth MAE ↓	PSNR ↑	Depth RMSE ↓	Depth MAE ↓
MultiMAE-B (RGB+D)	24.7	**280**	**184**	14.4	**215**	150
CROPCODE PC+RGB+D	**28.0**	423	245	**19.7**	216	**145**
CROPCODE four streams	26.9	483	299	18.3	263	183
CROPCODE headline	23.4	400	229	18.1	280	200
Mean visible plant depth	–	336	258	–	236	174
Cloud block	Chamfer ↓	F1@0.01 ↑	F1@0.02 ↑	Chamfer ↓	F1@0.01 ↑	F1@0.02 ↑
Point-MAE (PC)	**1.14**	**0.257**	**0.645**	**0.67**	**0.516**	**0.826**
CROPCODE PC only	9.58	0.044	0.225	4.95	0.061	0.230
CROPCODE PC+RGB+D	7.73	0.069	0.250	4.13	0.080	0.276
CROPCODE four streams	8.50	0.067	0.251	4.57	0.071	0.267
CROPCODE headline	5.93	0.134	0.351	2.94	0.161	0.431

Table 14. **Masked reconstruction against published encoders (validation, 2,250 plants per species).** Image block: 75% of the photograph and depth patches hidden, the point cloud hidden; PSNR (dB) on the plant pixels of the hidden patches; depth RMSE and MAE in mm on the same pixels. The last row fills hidden plant pixels with the mean depth of the visible ones (300 plants). Maize depth images are encoded per plant, so maize millimeters are approximate. Cloud block: the point cloud alone, 60% of its groups hidden; Chamfer ($\times 10^{-3}$, unit sphere) and F1 at two thresholds at 2,048 points per side. Point-MAE keeps the visible 40% of the points and is given the hidden groups' centers; CROPCODE is given neither. The CROPCODE encoders are those of Table 3 (epoch 600; headline epoch 1,000).

$\tau = 0.01$ is nearer Point-MAE's than its recall (sorghum 0.22 against 0.32, recall 0.10 against 0.22): it places points on the plant but misses part of it.

F.10. Latent space structure analysis (E9)

Frozen encoder `[CLS]` embedding spaces form continuous, smooth manifolds without discrete clustering, as expected for continuous synthetic generators (best silhouette coefficient 0.15–0.24 at $k = 2$, adjusted Rand index ≈ 0.0 against shape deciles, t-SNE trustworthiness 0.95–0.99). The latent effective participation ratio is low: 4.4–6.3 (out of 768 dimensions) in sorghum and 6.7–11.3 in maize. Explaining 90% of latent variance requires 14–15 principal components in sorghum and 18–19 in maize. In sorghum, the leading principal component aligns with global plant scale (correlation with leaf count $|r| = 0.73$–0.94).

F.11. Source of Occlusion

Rendering realistic occluded scenes is expensive, so we ask whether cheaper, synthetic occlusion can replace it during training. We compare four sources of occlusion (Fig. 22) and test every model on the same rendered occluded scenes.

Setup. Each rendered maize row shows a target plant with two neighbouring plants 0.15 m to either side, captured by a level camera 1 m away; the reconstruction target is the point cloud of the target plant alone, rendered from the same camera. All models are EmbodiedMAE-4M (base), start from the same checkpoint pretrained on single plants, and are trained for 200 epochs with a cosine schedule, learning rate 1.5×10^{-4}, a global batch of 32, 80% random token masking and the point-cloud loss QAL $+ 0.5\times$Sinkhorn. They differ only in the occlusion added to the training inputs. *Plant alone*: no occlusion. *Synthetic neighbours*: the point clouds of two other plants from the batch are placed beside the target in 3D, 0.15–0.35 m away, and rendered into the image and depth map with a z-buffer. *Bézier leaves*: three to eight leaf-shaped Bézier curves are drawn into the image and depth map, and their visible pixels are lifted to 3D points through the camera. *Rendered row*: the rendered scene itself. Half of the training samples receive the occlusion and half show the target plant alone, and the procedural parameters are hidden on half of the steps.

Table 15. Effect of the training occlusion on reconstruction of occluded maize plants. All models are tested on the rendered occluded rows (2,250 test plants, metrics at 10 mm).

Training occlusion	F1 ↑	Precision ↑	Recall ↑
Plant alone	0.416	0.460	0.391
Bézier leaves	0.552	0.639	0.507
Synthetic neighbours	0.717	0.764	0.691
Rendered row	**0.789**	**0.804**	**0.782**

Metrics. All models are tested on the rendered rows: 2,250 test plants, one fixed view each, with the occluded scene as input and no procedural parameters. Precision is the fraction of predicted points within 10 mm of the target plant, recall the fraction of target points with a prediction within 10 mm, and F1 their harmonic mean, averaged over plants.

Results. Tab. 15 and Fig. 23 show that the closer the training occlusion is to the test scenes, the better the reconstruction. Training on the plant alone gives the lowest F1 (0.416). Bézier leaves recover part of the gap (0.552), and synthetic neighbours, which are real plant geometry placed in 3D, recover most of it (0.717), but training on the rendered rows remains the best (0.789). Precision and recall improve together at every step.

F.12. Mask Ratio During Post-Training

We study how much of the input should be masked when the model is trained to reconstruct a plant that is partly hidden by its neighbours.

Setup. We use rendered sorghum rows: each scene shows a target plant with two neighbouring plants 0.15 m to either side, captured by a level camera 2.5 m away. The input is the RGB image, depth map and point cloud of the scene; the target is the point cloud of the target plant alone, rendered from the same camera. All variants are EmbodiedMAE-4M (base) and start from the same checkpoint, pretrained on single plants and then trained for 1,000 epochs on the rows at

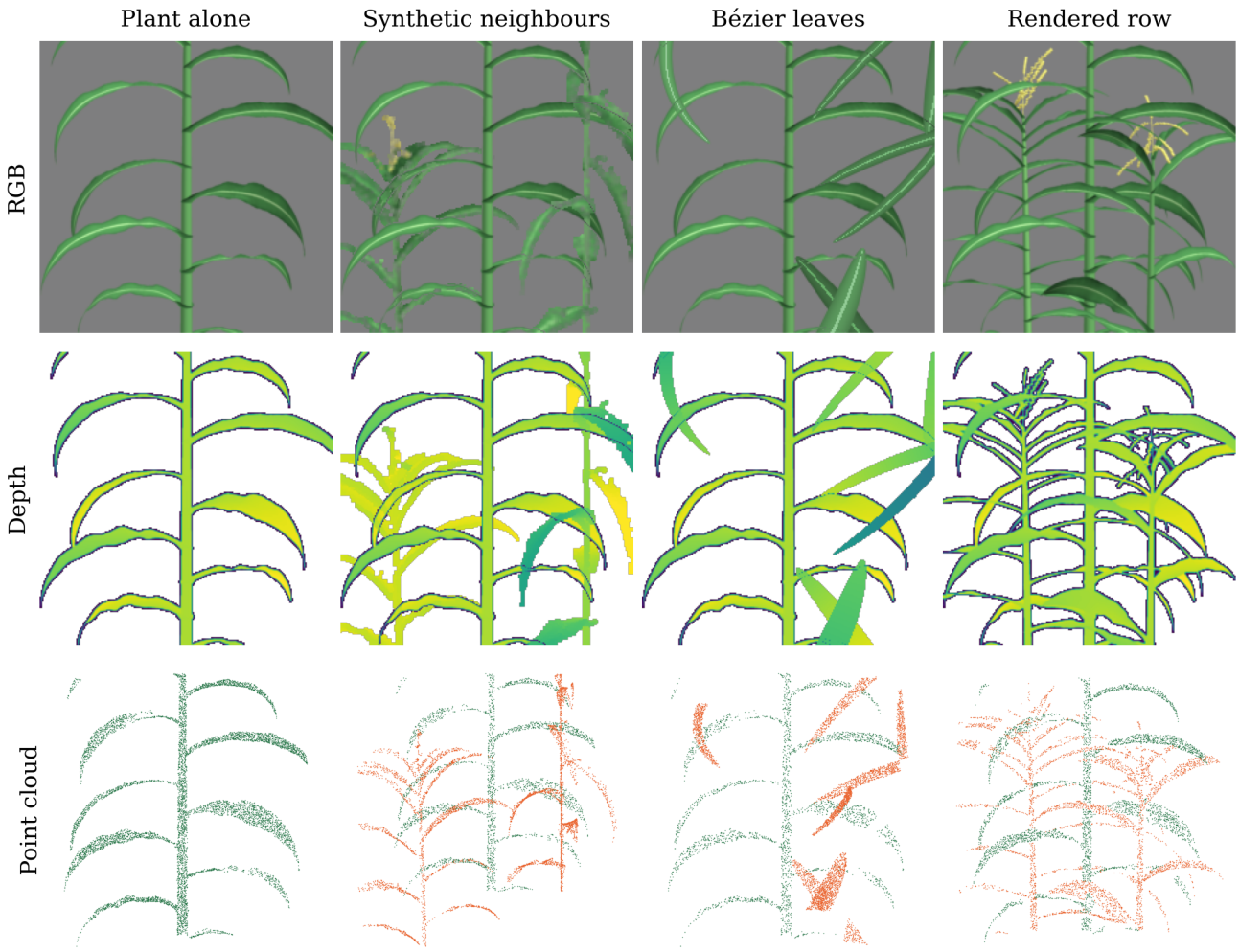


Figure 22. The four training sources of occlusion, shown for the same maize target plant. Columns: the target plant alone; synthetic neighbours (the 3D point clouds of other plants placed beside the target and rendered into the image and depth map); Bézier leaves (leaf-shaped occluders drawn in the image and depth map and lifted to 3D through the camera); and the rendered row. Rows: RGB image, depth map (one colour scale for all panels) and point cloud, with target points in green and occluder points in orange.

a mask ratio of 0.8. Each variant is fine-tuned for 200 further epochs at its own mask ratio m, with a cosine schedule, learning rate 5×10^{-5}, a global batch of 32 and the point-cloud loss QAL (10 mm) $+ 0.5\times$Sinkhorn. The mask ratio is the fraction of all input tokens that are hidden, split across the streams by a Dirichlet draw; $m = 0$ keeps every token visible. Half of the training samples show the occluded scene and half the target plant alone, and the procedural parameters are hidden on half of the steps. Since every variant starts from a model trained at $m = 0.8$, this ablation measures the mask ratio used for fine-tuning on occluded scenes.

Metrics. Each model is tested at the mask ratio it was trained with, on all 2,250 test plants (one fixed view each), with the occluded scene as input and no procedural parameters, using the same metrics at 10 mm as above.

Table 16. Effect of the mask ratio on occluded-plant reconstruction (rendered sorghum rows, 2,250 test plants, metrics at 10 mm). The mask ratio is the same at training and test time.

Mask ratio	F1 ↑	Precision ↑	Recall ↑
0.0	**0.576**	**0.606**	**0.551**
0.2	0.555	0.590	0.526
0.4	0.527	0.568	0.494
0.6	0.475	0.529	0.436
0.8	0.379	0.464	0.325

Results. Tab. 16 and Fig. 24 show that reconstruction im-

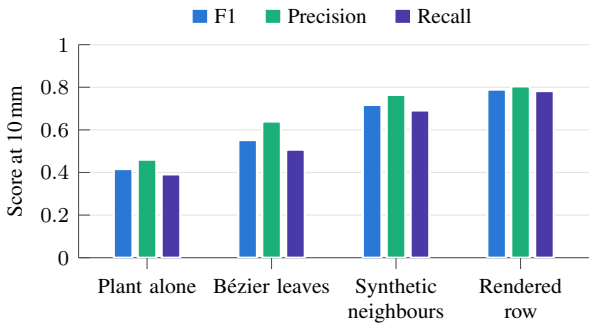


Figure 23. Reconstruction of occluded maize plants by training occlusion (data of Tab. 15). F1, precision and recall rise together as the training occlusion becomes more realistic.

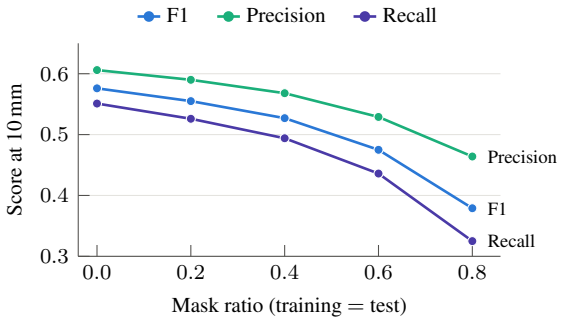


Figure 24. Occluded-plant reconstruction against the mask ratio (data of Tab. 16). All three metrics fall steadily as more of the input is hidden; keeping every token visible is best.

proves steadily as the mask ratio decreases: F1 rises from 0.379 at $m = 0.8$ to 0.576 with no masking, and precision and recall improve together. Note that every variant has already been trained with heavy masking (masked pretraining on single plants and 1,000 epochs on the rows at $m = 0.8$), so the benefit of masking for learning the representation is already in place. In this fine-tuning stage, hiding further input tokens only removes evidence about the target plant, and keeping the whole occluded scene visible works best. This does not imply that training without masking from scratch would be better, which we did not test.

F.13. Point-Cloud Loss

We compare point-cloud losses for reconstructing single sorghum plants.

Losses. Let $\mathcal{P}$ be the predicted point cloud and $\mathcal{T}$ the target cloud, and let

$$d(x, \mathcal{S}) = \min_{s \in \mathcal{S}} \|x - s\|_2 = \min_{s \in \mathcal{S}} \sqrt{\textstyle\sum_{k=1}^{3} (x_k - s_k)^2} \quad (8)$$

be the Euclidean (not squared) distance from a point x to the nearest point of a cloud $\mathcal{S}$. The quality-aware loss (QAL) weights every nearest-neighbour distance by how far it is from a tolerance τ:

$$\mathcal{L}_{\mathrm{QAL}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} w\big(d(p, \mathcal{T})\big)\, d(p, \mathcal{T}) + \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} w\big(d(t, \mathcal{P})\big)\, d(t, \mathcal{P}), \quad (9)$$

$$w(d) = \sigma\big(\alpha\,(d - \tau)\big), \quad (10)$$

where σ is the logistic function. Points within τ of the other cloud get a weight below $1/2$ (down to $\sigma(-\alpha\tau)$ at $d = 0$), while points far beyond τ get a weight close to 1, so training concentrates on the badly placed points, such as missing leaf tips; α sets how sharply the weight switches at τ. We use $\alpha = 100$ and $\tau = 10$ mm (the clouds are normalised per plant, so τ is 0.01 of the plant radius, *i.e.* 10 mm for a plant of median size). *Sinkhorn* is the debiased Sinkhorn divergence $S_\varepsilon(\mathcal{P}, \mathcal{T})$ ($p = 2$, blur 0.02) on 1,024 sampled points, and *QAL + Sinkhorn* is $\mathcal{L}_{\mathrm{QAL}} + \lambda\, S_\varepsilon(\mathcal{P}, \mathcal{T})$ with $\lambda = 0.5$. Each point-cloud loss enters the total loss with weight 1.

Table 17. Point-cloud losses on single plants (sorghum, trained from scratch, 2,250 test plants).

Loss	F1@5	F1@10	Prec.@10	Rec.@10
QAL	0.057	0.224	**0.561**	0.144
Sinkhorn	0.045	0.168	0.142	**0.206**
QAL + Sinkhorn	**0.083**	**0.259**	0.422	0.189

Single plants. Models are trained from scratch on Sorghum_15K (10,500 training plants, one random view per plant per epoch, each plant rendered alone) for 200 epochs (65.8k steps) with a cosine schedule, learning rate 1.5×10^{-4}, a global batch of 32 and 80% token masking, and differ only in the point-cloud loss. They are tested on all 2,250 test plants with the plant alone as input.

Metrics. Precision is the fraction of predicted points within a threshold of the target plant, recall the fraction of target points with a prediction within the threshold, and F1 their harmonic mean, averaged over plants. We write F1@x, Prec.@x and Rec.@x for a threshold of x mm.

Results. Neither kind of loss is sufficient on its own (Tab. 17). QAL, which matches every point to its nearest neighbour, places points precisely but leaves much of the plant uncovered (Rec.@10 0.144), while the Sinkhorn divergence covers more of the plant but places points poorly (Prec.@10 0.142). Adding a Sinkhorn term to QAL gives the best F1 at both thresholds: it improves F1@5 over QAL by 0.026 and F1@10 by 0.035 (better on 82% of plants), mainly by raising recall at a moderate cost in precision.

Table 18. QAL parameter screen (sorghum single plants, epoch 300, 225 validation views). Here @10 is a threshold of 0.01 of the plant radius (≈10 mm). The row $\tau = 10$ mm, $\alpha = 100$ is taken from a longer run (300k-step schedule) at the same epoch, so its learning rate had not yet decayed.

τ	α	F1@10	Prec.@10	Rec.@10
5 mm	100	0.155	0.398	0.107
10 mm	100	0.154	0.398	0.107
20 mm	100	0.139	0.384	0.097
10 mm	30	0.146	0.403	0.101
10 mm	300	**0.164**	**0.406**	**0.115**

QAL parameters. We screened the tolerance τ and the sharpness α of Eq. (10) in a shorter setting: sorghum single plants, 3,000 training plants, trained from scratch with a 30k-step (320-epoch) cosine schedule, 75% token masking and synthetic leaf occluders with sensor noise added to the training inputs, using QAL alone as the point-cloud loss. Each setting was trained once and is scored at epoch 300 on 225 clean validation views (Tab. 18).

A sharper weighting ($\alpha = 300$) gives the highest F1, precision and recall, and a tolerance of 20 mm is the weakest; a tolerance of 5 or 10 mm performs the same. The differences are small (at most 0.025 F1), come from a single seed and a short schedule, and should be read as a screen rather than a final ranking.

F.14. Mixed Masking

When the training input is occluded, the masking can target the occluder. *Structured* masking hides the tokens that show an occluder first (an image patch that is at least 30% occluder pixels, or a point-cloud token whose centre is an occluder point) and fills the rest of the masking budget at random. *Mixed* masking flips one coin per occluded training scene: with probability 0.5 the scene is masked structured, otherwise uniformly at random (Fig. 25). We compare mixed masking with uniform *random* masking. Validation and test always use random masking.

Setup. All models are EmbodiedMAE-4M (base) trained for 200 epochs with a cosine schedule, a global batch of 32, 80% token masking and the point-cloud loss QAL + 0.5×Sinkhorn; half of the training samples are occluded. The two models of each pair differ only in the masking. *Synthetic scenes*: the point clouds of two other plants from the batch are placed beside the target plant in 3D and rendered into its image and depth map; these models are trained from scratch and receive the target's procedural parameters as input in training and test. *Rendered rows*: the target plant is rendered with its two neighbouring plants 0.15 m to either side, and no procedural parameters are given at test. The maize rendered-row models are trained from scratch; the sorghum ones are fine-tuned from a model already trained for 1,000 epochs on the rendered rows with random masking. In the maize pair, the mixed model was trained with 8 samples per GPU on 4 GPUs and the random model with 16 per GPU on 2 GPUs (the same global batch).

Metrics. Every model is tested on all 2,250 test plants, one view each, with the occluded input. F1@x is the harmonic mean of precision (predicted points within x mm of the target plant) and recall (target points with a prediction within x mm), averaged over plants.

Results. Mixed masking helps only on the synthetic scenes: F1@10 rises by 0.048 in maize and by 0.013 in sorghum (Tab. 19). On the rendered rows it gives no gain in either species (-0.003 in maize and -0.004 in sorghum).

G. Real-Row Fine-Tuning Gallery (E10)

Figures 26 and 27 extend the real-row results of Sec. 4.8 to 20 more held-out plants, none of which the real-row fine-tune saw. Each tile shows the image the model reads, the true center plant as the camera sees it, and the center plant written before fine-tuning on real plants (the row fine-tune, trained on synthetic rows only) and after it (the real-row fine-tune, trained on the other 75% of the real plants). All three streams are in. Chamfer is in 10^{-4} m^2. The maize tiles are chosen across the range of the fine-tuned model's results, at the 10th, 30th, 50th, 70th, 90th and 98th percentile of its Chamfer in each row layout, so the bottom tiles are among its worst cases. Every tile improves: the maize Chamfer goes from 20–195 before fine-tuning to 6–52 after, and the predicted height moves from about 1.0 m for every plant to within a few centimeters of the true height (for example 1.52 against 1.52 m and 1.88 against 1.85 m). The worst maize tile after fine-tuning (Chamfer 52) also has the largest height error in the gallery, 1.23 against 1.30 m. The sorghum tiles cover all eight held-out plants. Their Chamfer goes from 661–2,781 before to 52–122 after; the fine-tuned model recovers the stalk over the full height of plants 2.8–4.9 m tall, but few of the leaves.

H. Row Occlusion Gallery (Regime III)

Figures 28 and 29 extend Fig. 5 to ten more held-out test plants per species. Unlike Fig. 5, these plants are not selected for clarity: they sit at the 5th, 15th, . . . , 95th percentile of the row-trained model's Chamfer distance (mean of the two yaws) over the 2,250 test plants of each species, so the lower rows show its harder cases.

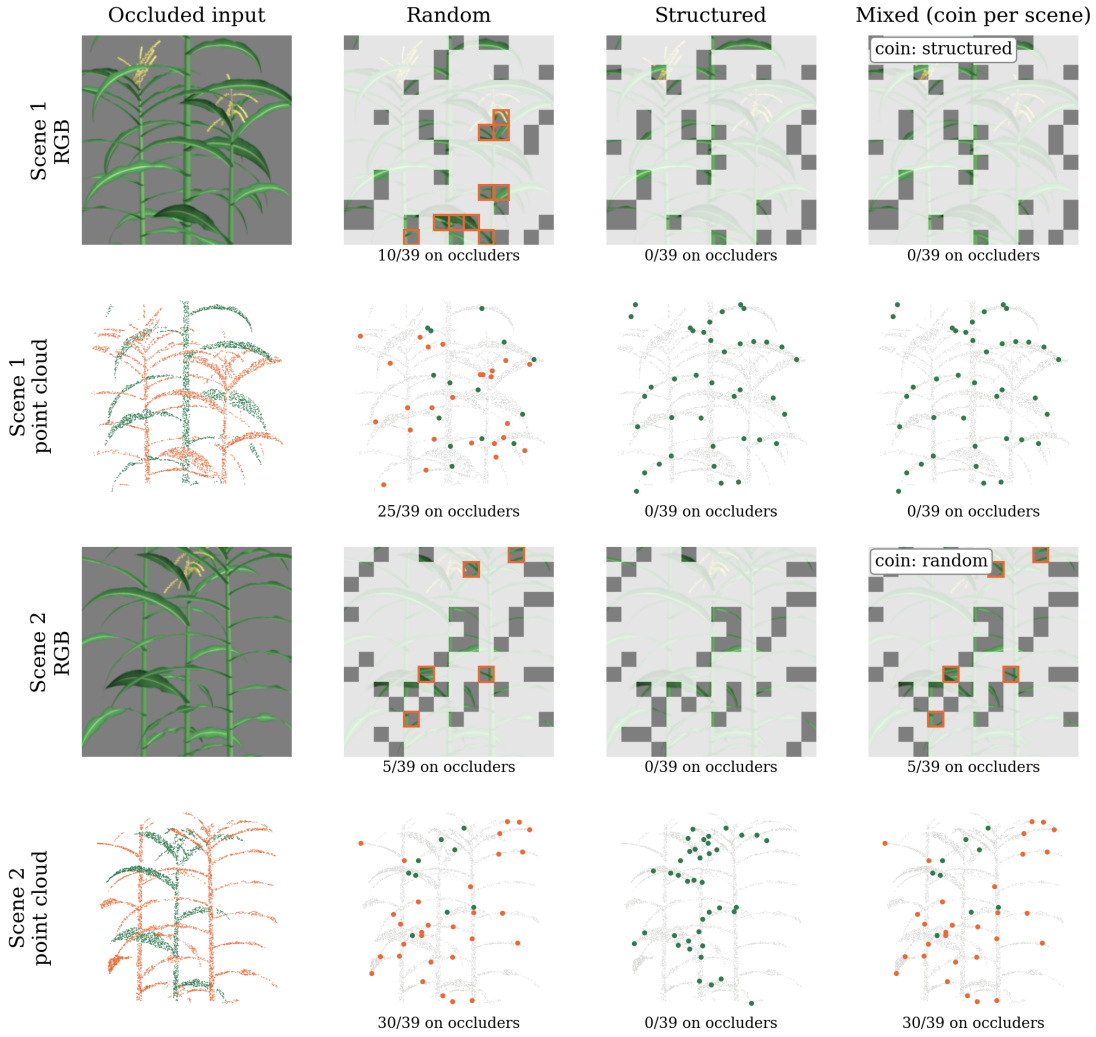


Figure 25. Mixed masking with one coin per scene, on two occluded maize scenes from the rendered rows. Columns: the occluded input (point cloud: target plant in green, occluders in orange); random masking; structured masking, which masks the occluder tokens first; and mixed masking, where a coin per training scene selects structured masking (scene 1) or random masking (scene 2). For illustration, every masked panel keeps the same 20% of the tokens of each modality visible, drawn with the same random numbers, so only the policy differs. Images: masked patches are faded, visible patches are in full colour (the render background is grey), and visible occluder patches are outlined in orange. Point clouds: dots mark the centres of the visible point tokens (green: target, orange: occluder). Below each panel: visible tokens that lie on occluders, out of 39.

Table 19. Mixed masking (one coin per scene) against random masking on occluded test input (2,250 test plants). Δ is mixed minus random in F1@10. The better F1@10 of each pair is in bold.

			Random		Mixed		
Training occlusion	Tested on	Species	F1@5	F1@10	F1@5	F1@10	ΔF1@10
Synthetic scenes	synthetic scenes	maize	0.160	0.439	0.190	**0.487**	+0.048
Synthetic scenes	synthetic scenes	sorghum	0.112	0.310	0.119	**0.323**	+0.013
Rendered rows	rendered rows	maize	0.525	**0.775**	0.522	0.771	−0.003
Rendered rows	rendered rows	sorghum	0.133	**0.377**	0.133	0.373	−0.004

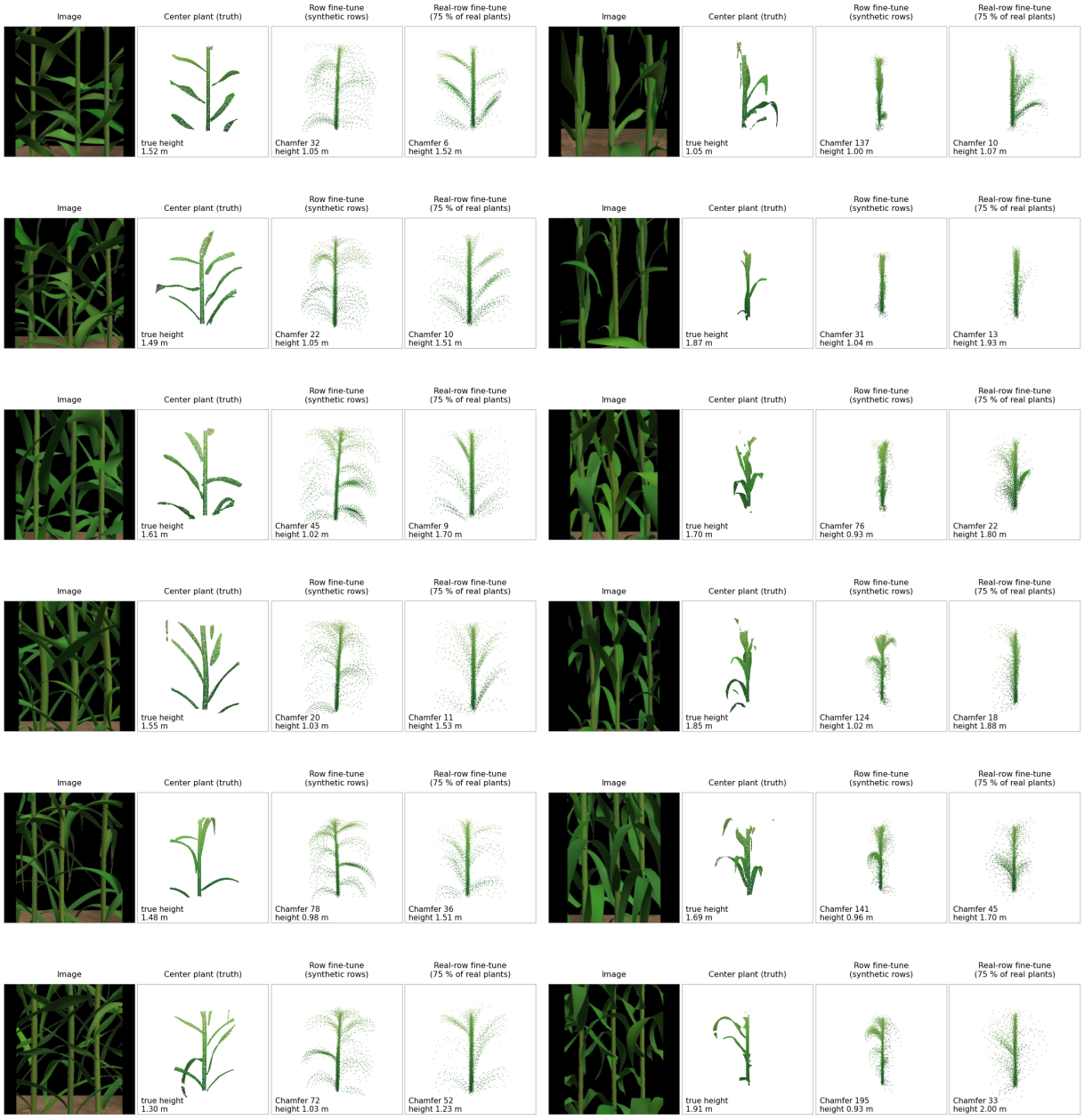


Figure 26. **Real maize rows, held-out plants (E10).** Left column: parallel rows; right column: perpendicular rows; from top to bottom the 10th, 30th, 50th, 70th, 90th and 98th percentile of the real-row fine-tune's Chamfer. In each tile: the image the model reads, the true center plant, and the center plant from the row fine-tune (synthetic rows) and from the real-row fine-tune (75% of the real plants), with Chamfer (10^{-4} m^2) and height.



Figure 27. **Real sorghum rows, all eight held-out plants (E10).** The middle camera stop of each plant, layouts alternating; columns as in Fig. 26. The image is the camera photograph, a tall strip framed to the whole plant. Sorghum has no height truth comparable to the recipe, so only Chamfer (10^{-4} m^2) is given.

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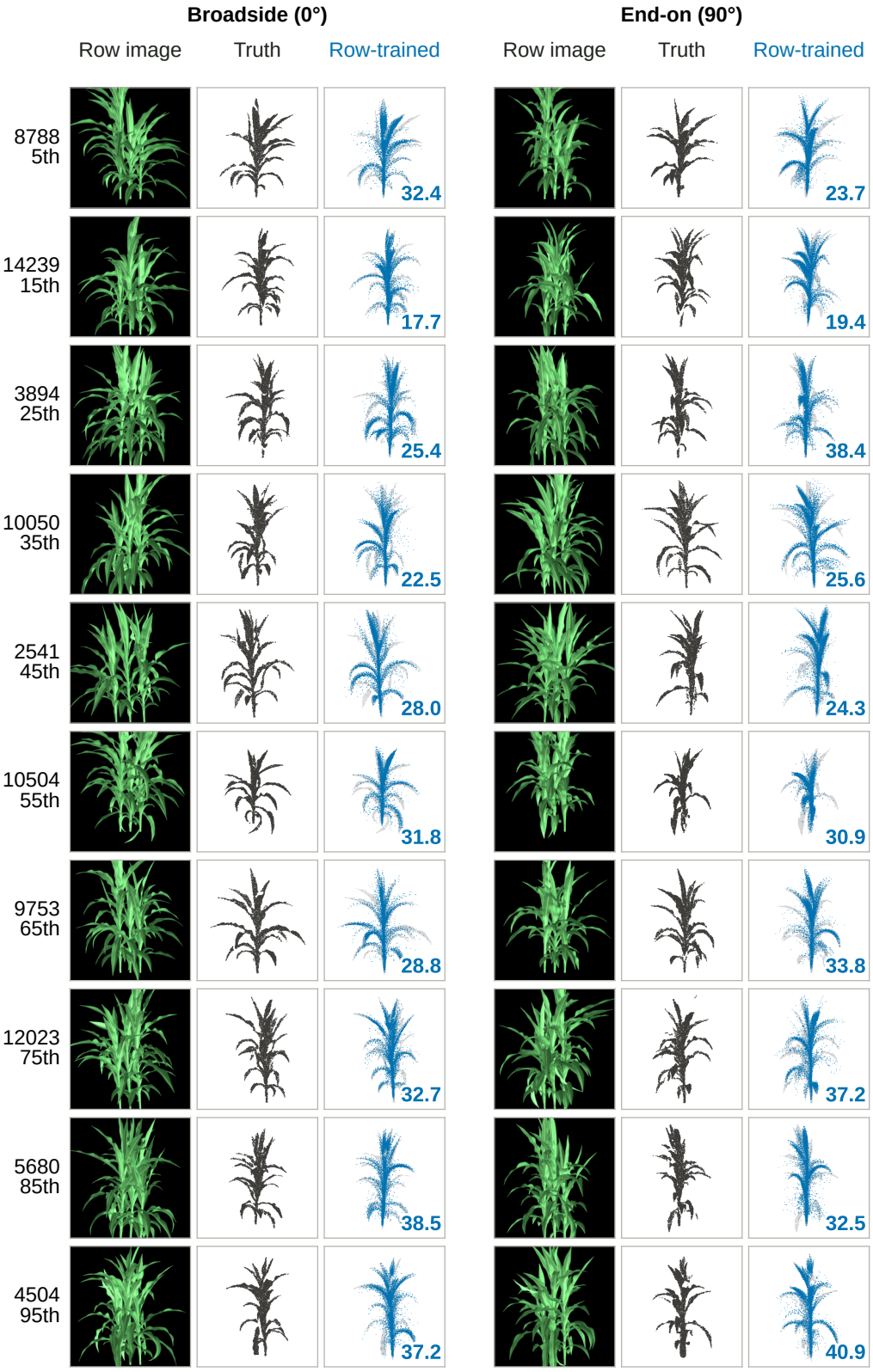


Figure 28. **Sorghum rows, ten held-out test plants at evenly spaced percentiles of the row-trained error.** Each row gives the plant number and its percentile; broadside (0°, left) and end-on (90°, right); columns as in Fig. 5: row image, truth, row-trained model (blue) over the truth in grey, Chamfer in 10^{-4} m^2.

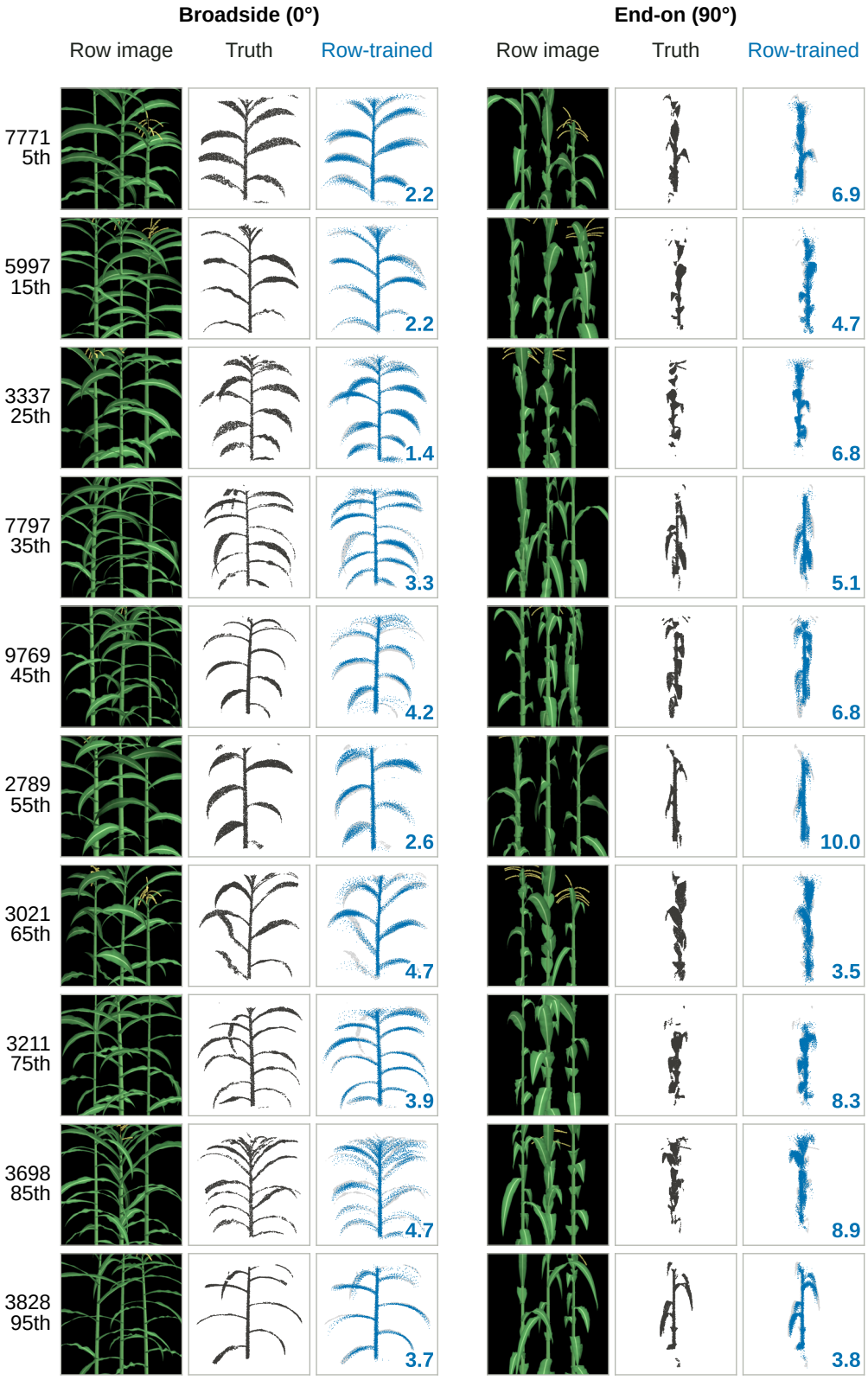


Figure 29. **Maize rows, ten held-out test plants at evenly spaced percentiles of the row-trained error.** Each row gives the plant number and its percentile; broadside (0°, left) and end-on (90°, right); columns as in Fig. 5: row image, truth, row-trained model (blue) over the truth in grey, Chamfer in 10^{-4} m^2.

I. Dataset Generation Details

FIELDSCAN contains four synthetic sets: isolated single-plant captures of maize and sorghum, alongside simulated *row scenes* captured by a level ground rover. All images are rendered at 1024×1024 resolution with a 40° vertical field of view ($f_x = f_y = 512/\tan 20^\circ = 1406.7$ px, principal point at image center) in OpenGL camera coordinates ($+y$ up, camera looking along $-z$).

I.1. Procedural crop generation

Maize procedural generation. Each plant i of 15,000 is generated from seed i using a procedural maize generator. Morphological traits are sampled from Gaussian distributions: plant height 0.832 ± 0.104 m, leaf count 12.35 ± 2.57, base leaf length 0.448 ± 0.057 m, base leaf width 0.066 ± 0.010 m, and base leaf insertion angle $43.98^\circ \pm 10.13^\circ$. Attributes are coupled through four latent traits (vigor, slenderness, posture, texture) with correlation strength 0.735, canopy position curves (bottom, middle, top), and per-leaf spatial jitter (scale 0.687). Plant heights span 0.62–1.44 m (median 1.01 m). Parameter records store plant height, leaf count, and per-leaf stem insertion height, length, width, azimuth, insertion angle, and droopiness. Per-leaf curvature is generated by constructing midrib B-splines and integrating bending curvature along leaf blades using 10-node Gauss–Legendre quadrature.

Sorghum procedural generation. Each plant i of 15,000 is generated from seed i using the EvoEngine sorghum generator. Heading angle is sampled uniformly in $[0^\circ, 360^\circ)$, stem tilt $0^\circ \pm 1.5^\circ$, leaf count 20 ± 2 (clamped to $[2, 128]$), and internode spacing 0.04 ± 0.03 m. Stem length is evaluated as internode $\times\, n_{\text{leaves}}/(1 - s_0)$, where s_0 represents the first leaf insertion point. Continuous curves over relative height $t \in [0, 1]$ determine leaf insertion height, blade length, width, roll ($\text{roll}_k = (k \bmod 2) \cdot 180^\circ + \text{roll}(t)$), branching angle, and bending curvature. Sorghum plants contain 4–35 leaves (mean 19.5) on stems ranging from 0.19 m to 1.74 m.

I.2. Dataset partitioning

Both species are partitioned into 10,500 training, 2,250 validation, and 2,250 test plants (70% / 15% / 15%).

- **Maize partitioning**: 19 scalar morphological features are robustly standardized using median absolute deviation ($z = (x - \text{median})/(1.4826 \cdot \text{MAD})$). Each plant is scored by Mahalanobis distance $\sqrt{z^\top (\Sigma + 10^{-6} I)^+ z}$. The most extreme plants are dealt alternately to test and validation splits to form an extreme tail split (train scores 1.56–4.60, validation 4.60–23.3, test 4.60–32.2).
- **Sorghum partitioning**: Plants are partitioned based on Mahalanobis distance across six core traits: leaf count, stem length, mean leaf length, maximum leaf length, mean leaf roll, and leaf roll standard deviation. Sampling probabilities scale continuously with extremeness rank, rising from 0.3% in the least extreme 5% of plants to 70% in the most extreme 5% (median rank 0.75 on validation and test, 0.37 on train). The rule is: each plant's extremeness is its Mahalanobis distance from the centroid of the six standardized traits, ranked to a percentile $r \in [0, 1]$; a pool of 30 % of the plants is drawn without replacement with probability proportional to $(r + 0.02)^2$ (seed 42), shuffled, and split equally into test and validation, and the rest is the training split. Re-running this procedure (`data_split/make_split.py`) reproduces the released assignment for all 15,000 plants.

Figure 30 shows the resulting split densities, the trait structure of the two generators and the occlusion the row scenes produce.

I.3. Isolated-plant sensor rendering

Off-screen rendering is executed via `pyrender`. Cameras target the plant bounding-box center $\mathbf{c}$ with world up $(0, 1, 0)$. Lighting consists of 0.15 ambient illumination plus three white directional lights directed at $\mathbf{c}$ from $(1, 1, 1)$, $(-1, 0.5, -0.5)$, and $(0, 1, 0)$ (intensities 4.0, 2.0, 1.5) against a neutral gray background (0.5). Depth $z \in [n, f]$ is packed losslessly into 32-bit RGBA PNG files:

$$v = \left\lfloor \text{clip}\left(\frac{z - n}{f - n}, 0, 1\right)(2^{32} - 1) \right\rfloor. \tag{11}$$

Pixels (u, v) with depth $z > 0$ back-project to 3D camera coordinates:

$$\mathbf{p}_{\text{cam}} = \left(\frac{u - c_x}{f_x} z,\ -\frac{v - c_y}{f_y} z,\ -z\right). \tag{12}$$

Maize captures. Ten camera directions per plant are sampled uniformly on the sphere (normalized Gaussian vectors), at a distance fitted to the plant: with r half the bounding-box diagonal,

$$\text{dist} = \frac{1.05\, r}{\sin 20^\circ}, \quad \begin{aligned} n &= \max(0.001,\ \text{dist} - 1.5\, r), \\ f &= \text{dist} + 1.5\, r. \end{aligned} \tag{13}$$

A fourth directional light (1.5) is mounted to the camera. Outputs include RGB images, RGBA images with cleared backgrounds, depth maps, camera poses, parameter records, and 8,192-point clouds back-projected from view depth maps.

Maize leaf curvature. The parameter record's per-leaf curvature is computed on the midrib B-spline rebuilt from the leaf's length, insertion angle, droopiness and spline point count (cubic, clamped, target elevation $\text{angle} + \text{droop} \cdot t^2$): over blade arc length $s \in [0.22, 1]$,

$$\theta = \int_{0.22}^{1} |\kappa(s)|\, ds, \qquad \bar{\kappa} = \theta / S, \tag{14}$$

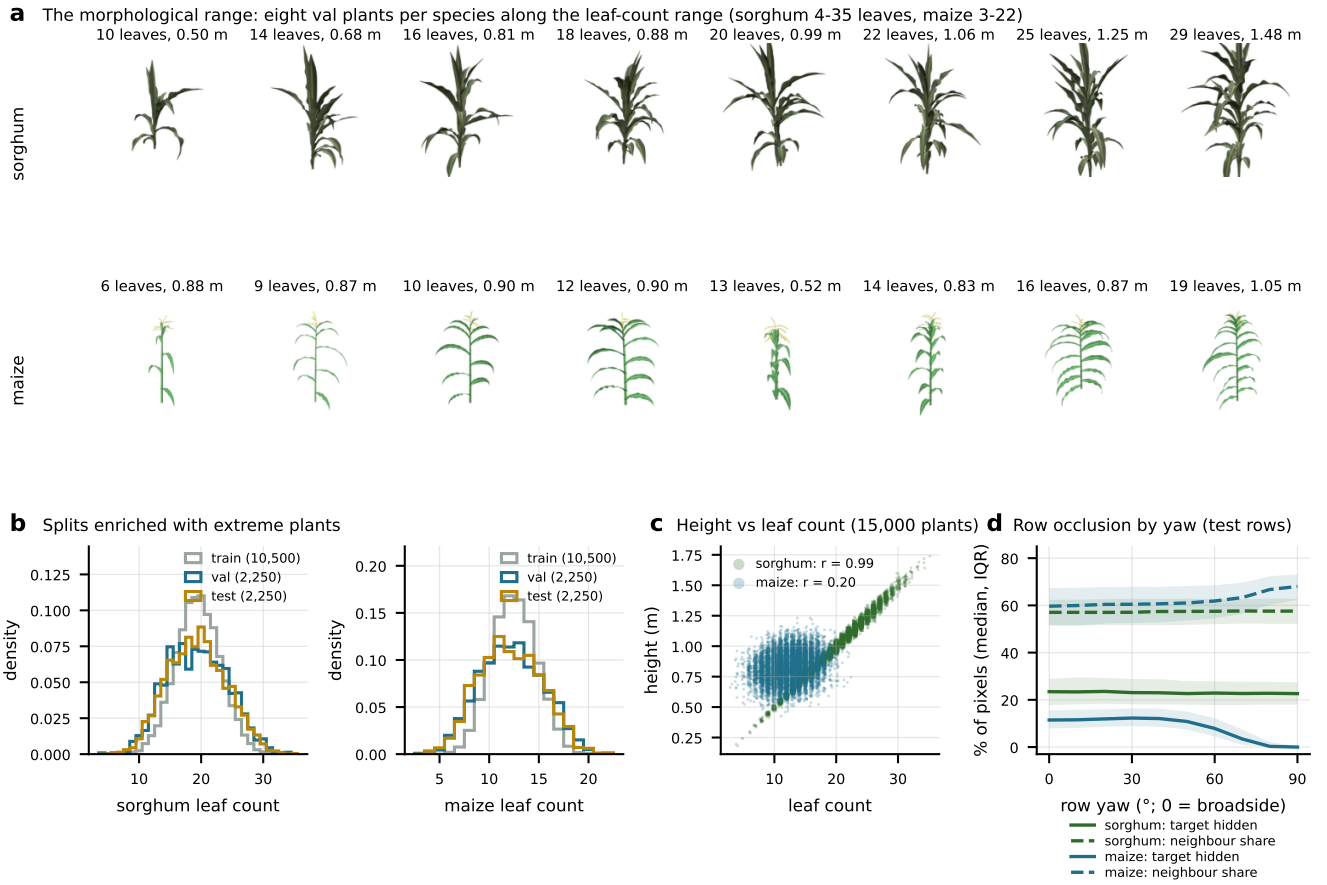


Figure 30. **What FieldScan spans.** (a) Eight validation plants per species along the leaf-count range, photographed near the horizon. (b) Leaf-count densities per split: validation and test are enriched with extreme plants. (c) Plant height against leaf count for all 15,000 plants per species: the sorghum generator ties the two ($r = 0.99$), in maize they are nearly independent ($r = 0.20$). (d) Row occlusion by yaw on the test rows: the share of the target's pixels hidden behind a neighbor (solid) and the share of plant pixels that belong to neighbors (dashed), median and interquartile range over plants.

with θ the total bend angle, S the blade length and $\bar{\kappa}$ the mean curvature, all integrated by 10-node Gauss–Legendre quadrature.

Sorghum captures. Ten camera directions form a fixed Fibonacci sphere rig at 2.5 m distance ($y_i = 1 - 2(i+0.5)/10$, $\phi_i = i(\sqrt{5}-1)\pi + \epsilon_i$ with a small fixed jitter ϵ_i; elevation angles $-64°$ to $+64°$, five above and five below the horizon). Outputs include RGB images, depth maps, camera poses, parameter records, and complete occlusion-free 8,192-point clouds expressed in the camera frame.

I.4. Simulated crop row rendering

Row composition. Within a split, row i places target main plant L_i between right neighbor L_{i+1} and left neighbor L_{i+2} at stem spacing $d = 0.15$ m along the x-axis. Rows are rendered across ten orientation angles $\theta \in \{0°, 10°, \ldots, 90°\}$ with random neighbor offsets $\delta \sim U(-15°, 15°)$. Sorghum neighbors are rotated so their stem axes run parallel to the main stem: each neighbor is turned about its stem base by the smallest rotation that makes its stem direction (the principal SVD axis of its stem vertices), after its own yaw, parallel to the main plant's stem direction after the main yaw. The main plant is never tilted, and maize is not tilted.

Physical contact solver. Inter-plant leaf collisions are resolved using an articulated contact solver. Leaf blades are modeled as chains of 8 cm segments connected by joints that bend along circular arcs, up/down and sideways. Stems, sheaths (within 2.5 cm of stem axes), and maize tassels remain rigid. Upon collision, joints rotate by the minimum clearing angle required to eliminate triangle–triangle intersection. Neighbors bend first; main plant leaves bend only when neighbors reach rotation limits (Tab. 20). When

no valid layout exists under default limits, limits loosen in stages, and stem spacing widens in 5 cm steps (refined to 1 cm); the maximum spacing used in maize is 0.33 m.

Stage	Collar side	Blade side	Up/down	Total turn
Stage 0 (default)	30°	10°	45°	90°
Stage 1 (medium)	60°	10°	60°	120°
Stage 2 (loose)	90°	30°	70°	180°

Table 20. **Physical contact solver staged bending thresholds.** Joint rotation limits applied across optimization stages to resolve inter-plant row leaf collisions.

Contact detection (MEETS). Whether a leaf *meets* another plant is decided in two steps. (1) *Grid screen*: each plant marks a 1 cm occupancy grid at surface samples that cover every surface point to within $\varepsilon_O = 2.8$ mm, and the marked cells are grown by one cell in every direction; the tested leaf is sampled so that every point of it lies within $\varepsilon_M = 6.8$ mm of a sample. Where two surfaces cross, some tested sample and some marked sample lie within $\varepsilon_M + \varepsilon_O < 1$ cm of each other, so the tested sample falls in a grown cell: no tested sample in another plant's grown cells means no crossing. (2) *Exact test*: each triangle with a sample in a grown cell is tested exactly, triangle against triangle, against that plant as it currently stands. Only an actual intersection counts as a contact, so leaves may rest against each other with no margin. Rigid parts (stems, sheaths, the maize tassel) are tested by the same exact triangle tests; the tassel is kept out of the grids, since its thin branches can interleave with a neighbor's without crossing. Two surfaces *cross* when a triangle of one plant intersects a triangle of another; they *touch* when they meet without intersecting. Crossing is forbidden, touching is allowed.

Bending a leaf clear. Algorithm 1 finds the smallest change of a leaf's shape that clears a set of obstacles. Candidate bends at a joint are up/down angles in 5° steps and sideways angles in 10° steps within the stage's limits (Tab. 20), ordered by magnitude, sideways bends last among equals, down before up; three halvings toward the joint's previous bend then find where the leaf just starts to clear, so it comes to rest on or under what it meets. The search runs within a budget of segment tests (12,000 per leaf; 50,000 in stage 2). For sorghum neighbors, a leaf with no clear bend may also *roll* about its stem axis, as if it had grown at another azimuth: rolls of $\pm 20°, \pm 40°, \pm 60°, \pm 80°$ are tried nearest first, each followed by the same bending search. Rolled neighbor leaves are exempt from the own-plant rule (a leaf at a new azimuth may cross its own siblings, as a leaf of any sampled plant could); maize leaves do not roll, and main-plant leaves never roll.

Settling (resting against). The grid screen can leave a gap of up to about 2 cm under a bent leaf. Once a layout is solved, every bent leaf eases back toward its grown shape: all its joint bends are scaled down together, and 12 halvings of the scale find the least bend (to $1/4096$ of the bend) that still clears every other plant by exact triangle tests on the positions as rendered. The leaf then rests against what it bent round, a fraction of a millimetre away; a leaf whose grown shape clears by then returns to it. Three passes are made, since a leaf that settled can make room for one settled before it.

Solving a layout and choosing the spacing. Algorithm 2 resolves all contacts of one layout under one stage's limits: stems and sheaths never move (if they cross, the layout fails at once); the main plant bends first, but only against the neighbors' stems and sheaths, which cannot give way; the neighbors then give way to the main plant and to each other in rounds; and only if a neighbor leaf is boxed in do the main-plant leaves in its way bend round it. The main plant therefore keeps its grown, ground-truth shape wherever a neighbor can get out of its way. Algorithm 3 wraps the stages and the spacing search: the solver first tries $d = 0.15$ m, raises the spacing in 5 cm steps up to 1.5 m until a stage succeeds, then refines in 1 cm steps from the last failure toward that success. Whether bending succeeds is not monotone in d, so this finds the first working spacing along that search, not necessarily the smallest. The spacing used, the stage, the number of bent and rolled leaves per plant, the largest leaf turn and the solve time are recorded per orientation.

Independent mesh intersection validation. Solved row layouts are verified by an independent validation suite that shares no code with the solver: it executes exact triangle–triangle intersection checks between adjacent plant meshes at their rendered positions, and a row passes only if no triangle of one plant intersects a triangle of another. The check found no inter-plant mesh intersections on the rows it was run on; which rows those were is not recorded, and the bent meshes are not stored with the renders, so it cannot be repeated on the released data. The solver's own record covers every row: across all 300,000 row orientations (15,000 rows × 10 yaws per species), none is marked failed, and the spacing had to be widened beyond 0.15 m in 21,599 sorghum (14.4 %) and 34,388 maize (22.9 %) orientations.

Rover camera capture. Row perspectives are captured by a ground rover camera positioned at $(0, h, \mathrm{dist})$ aimed horizontally at $(0, h, 0)$, where h represents target plant bounding-box center height ($\mathrm{dist} = 1.0$ m for maize, 2.5 m for sorghum). Each shot is rendered twice: (1) full row scene input (RGBA, depth, and 24,576-point scene cloud

Algorithm 1 BENDCLEAR: the smallest clear shape of one leaf

Require: leaf ℓ with segments $0, \dots, n-1$ (collar first); obstacles $\mathcal{O}$; stage limits (U, S_0, S, T)

$p \leftarrow$ grown shape (every joint bend $(\alpha_j, \beta_j) = (0, 0)$)
$j \leftarrow$ first segment of $\ell(p)$ that MEETS $\mathcal{O}$ ▷ none: clear
while j exists **do**
  if too many steps or test budget spent **then return** FAIL
  $q^\star \leftarrow$ none
  for $k = j, j-1, \dots, \max(0, j-3)$ **do** ▷ joint before the contact first
    for all bends (α, β) of joint k, smallest first **do**
      $q \leftarrow p$ with joint k set to (α, β), joints after k straightened
      if $\sum_i \|(\alpha_i, \beta_i)\| > T$ **or** segs. $k..j$ of $\ell(q)$ MEETS $\mathcal{O}$ **then**
        continue
      $q \leftarrow$ least bend between p and q at joint k that clears (3 halvings)
      if $\ell(q)$ enters own stem or crosses own plant more than as grown **then**
        continue
      $q^\star \leftarrow q$; **break** both loops
  if $q^\star$ is none **then return** FAIL
  $p \leftarrow q^\star$; $j \leftarrow$ first segment $\geq k$ of $\ell(p)$ that MEETS $\mathcal{O}$
return p

For a sorghum neighbor leaf, on FAIL the search is repeated from the grown shape rolled about the stem by $\pm 20°, \pm 40°, \dots$ up to $80°$; the first roll that yields a clear shape is kept.

Algorithm 2 SOLVELAYOUT: contacts of one layout under one stage

Require: main plant P_0, neighbors P_1, P_2 at spacing d, angles θ; stage limits

if stems or sheaths of any two plants cross **then return** FAIL
for all leaves ℓ of P_0 that MEETS the stems/sheaths of P_1, P_2 **do**
  $\ell \leftarrow$ BENDCLEAR(ℓ, stems and sheaths of P_1, P_2)
  if FAIL **return** FAIL
for round $= 1, \dots, 8$ **do** ▷ neighbors give way
  for $r = 1, 2$ **do**
    for all leaves ℓ of P_r that MEETS the other two plants **do**
      $\ell \leftarrow$ BENDCLEAR(ℓ, other two plants); on FAIL mark ℓ stuck
  if no leaf changed **then break**
for $t = 1, 2$ **while** some neighbor leaf is stuck **do** ▷ main gives way only where it must
  bend clear the P_0 leaves that MEETS a stuck leaf; rerun the neighbor rounds
return OK if no triangle of any plant crosses another plant (exact test), else FAIL

Algorithm 3 SOLVEORIENTATION: spacing, stage and settling

Require: plants P_0, P_1, P_2; angles θ; $d_0 = 0.15$ m, $d_{\max} = 1.5$ m

function TRY(d)
  for stage $s = 0, 1, 2$ **do**
    $r \leftarrow$ SOLVELAYOUT($P_0, P_1, P_2, d, \theta, \text{limits}_s$)
    if $r =$ OK **or** stems/sheaths cross **then return** r
  return FAIL
if TRY(d_0) = OK **then** $d \leftarrow d_0$
else
  $d \leftarrow$ first of $d_0{+}0.05, d_0{+}0.10, \dots, d_{\max}$ with TRY(d) = OK
  $d_{\text{lo}} \leftarrow$ the last spacing tried before d (a failure)
  $d \leftarrow$ first of $d_{\text{lo}}{+}0.01, d_{\text{lo}}{+}0.02, \dots < d$ with TRY = OK, else keep d
settle every bent leaf of the layout kept at d (resting against)
render the input (three plants) and the ground truth (main plant alone, same bent shape)

Split	Orient.	At d=0.15 m	Stage 0 / 1 / 2	Max / p99 d	Bent	Rolled
Maize train	105,000	76.4%	81,679 / 16,138 / 7,183	0.33 / 0.23 m	48.0%	–
Maize val	22,500	78.3%	17,780 / 3,321 / 1,399	0.28 / 0.23 m	46.8%	–
Maize test	22,500	79.0%	17,824 / 3,284 / 1,392	0.30 / 0.23 m	47.1%	–
Sorghum train	105,000	85.4%	42,272 / 25,120 / 37,608	0.39 / 0.21 m	98.3%	99.4%
Sorghum val	22,500	86.0%	9,395 / 5,335 / 7,770	0.63 / 0.21 m	97.8%	98.9%
Sorghum test	22,500	86.3%	9,511 / 5,416 / 7,573	0.46 / 0.21 m	97.3%	99.1%

Table 21. **Crop row solver statistics.** Solution convergence rates across solver stages and required plant spacing across splits. "Bent" counts orientations in which at least one main-plant leaf was bent, "Rolled" those in which at least one neighbor leaf was rolled. No orientation failed in either species. Sorghum rows are more crowded than maize rows: in nearly every sorghum orientation some main-plant leaves are bent and some neighbor leaves are rolled, and about a third need the stage-2 limits, but most are still solved at the base spacing.

containing target and neighbor geometry), and (2) target main plant ground truth in its contact-bent shape (RGBA, depth, 8,192-point cloud, and procedural parameter record).

Caveats. The row ground truth is the main plant as bent in its row, while its parameter record describes the grown plant; main-plant leaves are bent only where unavoidable. At the 2.5 m sorghum distance, the $40°$ lens covers about 1.8 m of height, so the top or base of the tallest sorghum plants can leave the frame. Sorghum plant 0 uses seed 0, for which the generator does not reseed its random generator; its recipe matches its mesh where it has parameters (the leaf center lines lie a median 3.7 mm from the mesh, as for plants 1–40, 3.5–3.9 mm), but 2 of its 13 leaves carry no parameters, and the mesh is 0.35 m taller than its stem length predicts (about six standard deviations off the relation of the other plants). Plant 0 is one of the 10,500 training plants and in no evaluation split.